Finite-time breakdown for smooth compactly supported Euler data in every dimension $d \geq 3$

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Abstract

For every fixed integer $d \geq 3$ we construct smooth compactly supported, divergence-free initial data for the incompressible Euler equation on $\mathbb{R}^d$ whose maximal smooth Sobolev solution has finite lifetime. The Frobenius norm of its velocity gradient has infinite endpoint limsup, and the time integral of the L^∞ norm of $(\Omega_{ij})_{i<j}$ diverges. The pressure has the standard Newton normalization. The proof extends the localized oscillatory construction and scalar amplification mechanism introduced in [1]. The higher-dimensional argument retains the whole transverse space and the full pressure coupling, replaces curl by antisymmetric potentials, uses reflection only for the central endpoint selection, and proves a quantitative ambient relative estimate. A variational history solver, polynomial bounds for the actual lifted particle flow, and one fixed physical mean cutoff give a single frequency sequence and summability of the initial increments at every Sobolev order. No bound uniform in the dimension is required.

1 Statement and conventions

For each fixed integer $d \geq 3$, let $C^\infty_{c,\sigma}(\mathbb{R}^d)$ denote smooth compactly supported vector fields with zero divergence. All dimension-dependent constants, derivative orders, starting indices and scale thresholds may depend on this fixed d.

Theorem 1.1 (H_d)**.** *There exists $u_0 \in C^\infty_{c,\sigma}(\mathbb{R}^d)$ whose unique maximal smooth Sobolev solution of*

$$u_t + (u \cdot \nabla)u + \nabla P = 0, \qquad \operatorname{div} u = 0, \qquad u(0) = u_0$$

has $0 < T^*(u_0) < \infty$. Here the pressure is exactly

$$P = (-\Delta)^{-1} \sum_{i,j=1}^{d} \partial_i \partial_j (u_i u_j),$$

and, on every $[0, S]$ with $S < T^*$, the solution lies in $C([0, S]; H^m) \cap C^1([0, S]; H^{m-1})$ for every integer $m \geq 1$. With $\Omega_{ij} = \partial_i u_j - \partial_j u_i$, it satisfies

$$\limsup_{t \uparrow T^*} \operatorname*{ess\,sup}_x \left(\sum_{i,j} |\partial_i u_j(t, x)|^2 \right)^{1/2} = \infty, \qquad \int_0^{T^*} \operatorname*{ess\,sup}_x \left(\sum_{i<j} |\Omega_{ij}(t, x)|^2 \right)^{1/2} dt = \infty.$$

The three-dimensional conclusion is the theorem proved in [1], with the Newton pressure and smooth Sobolev class specified above. The present argument applies in that dimension with a transverse space of dimension two. For $d > 3$, every velocity and every localization is defined on $\mathbb{R}^d$. In particular, the construction retains compact support and finite energy in all spatial variables.

The scalar amplification equation, the variational history method, and the finite oscillatory expansion follow the structure of [1]. Their scalar and Hilbert-space estimates are reproduced below with their hypotheses. The dimension-dependent steps are the antisymmetric-potential localization, the full $(d-1)$-dimensional transverse evolution, its coupling to the pressure, and the Sobolev orders in the nonlinear correction and continuation argument. These steps are proved in the ambient dimension. The parameter estimates then provide one frequency sequence for which all initial Sobolev increments are summable.

Throughout, C_d denotes a positive constant depending on the fixed dimension and on fixed auxiliary cutoffs. Constants may change between estimates. Dependence on a derivative order is indicated explicitly. The notation $A \lesssim_d B$ means $A \le C_d B$, and $A \asymp_d B$ means both inequalities. The principal stage parameters are listed below; each is defined precisely at its first use.

$t_0,\ \tau$	physical activation time and scaled forward time
$\rho,\ k,\ \kappa = k^{-1}$	physical localization length and frequency parameters
$\delta,\ f_\delta$	angular width and periodic primary profile
P	fixed angular period
$h_*,\ h_{\mathrm{child}}$	inherited shear size and prescribed new shear size
$\Theta,\ e$	scaled time bound and geometric comparison error
$\alpha,\ A_{\mathrm{tar}}$	primary amplitude and unnormalized target size
$\mathcal{P}$	bound for the fixed source coefficients and inverse costs
$\ell_*,\ r_*$	fixed physical mean-cutoff parameters

2 The ambient equation and the endpoint criteria

Fix an integer $d \ge 3$. All constants in this paper may depend on d. We use the Euclidean norm on vectors, the operator norm on linear maps when estimating compositions, and the Frobenius norm in the statement of the singularity conclusions. Put $s = \lfloor d/2 \rfloor + 3$. In particular, $s > d/2 + 1$, and $H^s(\mathbb{R}^d)$ embeds into $C_b^{1,\alpha}$ for some $\alpha \in (0,1)$ fixed throughout this section. For smooth vector fields, $\|u\|_{H^m}^2 = \sum_{|\gamma|\le m} \|\partial^\gamma u\|_2^2$; replacing multi-indices by ordered words changes only finite constants.

Lemma 2.1 (Local theory and persistence)**.** *Every solenoidal datum in all integer Sobolev spaces has a unique smooth Sobolev Euler solution on an interval of length bounded below in terms of its H^s norm. The solution belongs to $C_t H^m \cap C_t^1 H^{m-1}$ on every shorter closed interval, for every $m \ge 1$. For $m \ge s$ it satisfies*

$$\frac{d}{dt}\|u\|_{H^m} \le C_{d,m} \|\nabla u\|_\infty \|u\|_{H^m}. \tag{1}$$

The same solution is obtained at every regularity level.

Proof. Let J_N be the orthogonal Fourier projection onto $\{|\xi| \le N\}$ and solve $\partial_t u_N = -J_N \Pi((u_N \cdot \nabla) u_N)$ with $u_N(0) = J_N u_0$, where Π is the solenoidal projection. The right-hand side is locally Lipschitz on the band-limited subspace. Both projections commute with differentiation; $J_N u_N = u_N$ and $\Pi u_N = u_N$. Consequently they disappear when the differentiated equation is tested against a derivative of u_N. For each differentiated transport term the top derivative cancels against its divergence-free transport field; the pressure cancels against a solenoidal test field. At order one the remaining integral is bounded directly by $C_d \|\nabla u\|_\infty \|\nabla u\|_2^2$. For $m \ge 2$ the remaining terms are products with derivative orders j and $m - j + 1$. The Gagliardo–Nirenberg inequalities with endpoints $\nabla u \in L^\infty$ and $D^m u \in L^2$ give, for $1 \le j \le m$,

$$\|D^j u\|_{L^{2(m-1)/(j-1)}} \le C_{d,m} \|\nabla u\|_\infty^{1-(j-1)/(m-1)} \|D^m u\|_2^{(j-1)/(m-1)}.$$

The exponent is interpreted as infinity at $j = 1$. The two reciprocal Lebesgue exponents sum to $1/2$, and the two interpolation parameters sum to one. This proves the homogeneous top-order estimate. Summing the lower orders and retaining conservation of L^2 energy proves (1). The argument uses coordinate differentiation and integration by parts, not a three-dimensional curl identity.

At order s, the Sobolev embedding gives $y' \le C_d y^2$. Thus the cutoff solutions have a common interval, depending only on the initial H^s norm. Every higher norm is bounded there by (1). The discarded frequencies of the quadratic term have L^2 norm tending to zero uniformly on shorter intervals, since that term is bounded in arbitrarily high Sobolev spaces. The L^2 difference energy estimate, followed by interpolation with the uniform higher bounds, yields convergence in $C_t H^m$ for every fixed m. The projected equation then gives convergence of time derivatives in $C_t H^{m-1}$. It also supplies the one-sided derivatives at the time endpoints. The same difference estimate proves uniqueness. Consequently the solutions produced using different cutoff orders or Sobolev indices agree on their common intervals. This proves persistence and defines a single maximal smooth Sobolev solution. □

Lemma 2.2 (Literal Newton pressure)**.** *For a smooth solenoidal u in all Sobolev spaces, set $F = \sum_{i,j} \partial_i \partial_j (u_i u_j)$ and $N_d(x) = |x|^{2-d}/((d-2)|\mathbb{S}^{d-1}|)$ for $x \ne 0$. Since $|\mathbb{S}^{d-1}| = d\,|B_1|$, this coefficient is exactly $[d(d-2)|B_1|]^{-1}$, as in the frozen normalization. The integral $P = N_d * F$ is absolutely convergent at every point and smooth. Its gradient is the L^2 gradient part of $-(u \cdot \nabla)u$. Thus the projected equation gives the Euler equation with exactly this pressure, without an undetermined function of time.*

Proof. Every derivative of F is both bounded and integrable: expand by the product rule and use the L^2 bounds for the two factors of each product. The kernel is locally integrable and bounded outside the unit ball, so the Newton integral is absolutely convergent. Equivalently, with $G_t(x) = (4\pi t)^{-d/2} e^{-|x|^2/(4t)}$,

$$N_d * F = \int_0^\infty G_t * F \, dt.$$

Indeed $\int_0^\infty G_t(x)\,dt = N_d(x)$ for $x \ne 0$, by the substitution $r = |x|^2/(4t)$ and the Gamma integral. Absolute Fubini follows from $\|G_t * |D^\gamma F|\|_\infty \le \min(\|D^\gamma F\|_\infty, C_d t^{-d/2} \|D^\gamma F\|_1)$. This envelope is time integrable because $d > 2$. It also permits differentiation under the integral to every spatial order.

For each coordinate derivative, the small-time L^2 bound is $\|\partial_i F\|_2$, while the large-time bound is $\|\partial_i G_t\|_2 \|F\|_1 \le C_d t^{-d/4-1/2} \|F\|_1$. The latter is integrable at infinity for $d > 2$. Hence $\nabla P \in L^2$, and the same argument applies to all its derivatives. The heat equation and its limits at zero and infinity give $-\Delta P = F$. Writing $B = (u \cdot \nabla)u$, incompressibility gives $\operatorname{div} B = F$, so $\nabla P + B$ is solenoidal. Moreover ∇P lies in the closed L^2 gradient space. Indeed its distributional curl vanishes. After Fourier transformation, its L^2 transform is parallel to ξ for almost every $\xi \ne 0$; this is exactly the range of the orthogonal gradient multiplier $\xi \otimes \xi/|\xi|^2$. The zero frequency is a set of measure zero. Cutting off in frequency away from zero and then approximating the potentials by compact smooth functions identifies this range with the closed L^2 gradient space. Orthogonal Helmholtz projection therefore yields $\nabla P = \Pi B - B$.

This normalization is also preserved under the limiting procedure in Lemma 2.1. More generally, suppose $u_n \to u$ in $C([0, S]; H^m)$ for every finite m, and define F_n, P_n by the same formulas. For each multi-index γ, the product rule, Cauchy–Schwarz, and Sobolev embedding give

$$\sup_{t \le S} (\|D^\gamma (F_n - F)\|_1 + \|D^\gamma (F_n - F)\|_\infty + \|\nabla D^\gamma (F_n - F)\|_2) \longrightarrow 0.$$

Splitting the heat integral at time one in the two estimates above yields

$$\|D^\gamma(P_n - P)\|_\infty \le C_d(\|D^\gamma(F_n - F)\|_\infty + \|D^\gamma(F_n - F)\|_1),$$
$$\|\nabla D^\gamma(P_n - P)\|_2 \le C_d(\|\nabla D^\gamma(F_n - F)\|_2 + \|D^\gamma(F_n - F)\|_1).$$

These estimates are uniform on $[0, S]$. Thus the Newton scalars converge with every spatial derivative uniformly, and their gradients converge in every Sobolev space. The same estimates applied to two nearby times give their time continuity. In particular the normalization is specified by the convolution before and after passage to the limit. $\square$

Lemma 2.3 (Stability on varying intervals)**.** *Let u be a smooth Euler solution on $[0, S]$ and let v_n be smooth solutions on $[0, S_n]$, where $0 < S_n \le S$. If $v_n(0) \to u(0)$ in H^s, then*

$$\sup_{0\le t\le S_n} \|v_n(t) - u(t)\|_{H^s} \longrightarrow 0.$$

No uniform higher norm of v_n is assumed.

Proof. Put $w = v_n - u$. Its equation is $w_t + (u + w) \cdot \nabla w + w \cdot \nabla u + \nabla q = 0$. Differentiation through order s, the transport cancellation, and the Sobolev product estimate give

$$y' \le C_d\|u\|_{H^{s+1}} y + C_d y^2, \qquad y = \|w\|_{H^s}.$$

All coefficients involving u are bounded on $[0, S]$ independently of n. While $y \le 1$, Gronwall gives $y(t) \le y(0)e^{CS}$. For large n this is less than $1/2$; continuity excludes a first time when $y = 1$. The estimate therefore holds on the whole actual interval $[0, S_n]$. Sobolev embedding also gives uniform convergence of the velocity gradients there. $\square$

Lemma 2.4 (Continuation and the exact vorticity norm)**.** *Let u be the maximal solution from Lemma 2.1, and suppose its maximal lifetime T^* is finite. Write $\Omega_{ij} = \partial_i u_j - \partial_j u_i$ and $|\Omega| = (\sum_{i<j} |\Omega_{ij}|^2)^{1/2}$. Then*

$$\limsup_{t\uparrow T^*} \|\nabla u(t)\|_{L^\infty,\mathrm{F}} = \infty, \qquad \int_0^{T^*} \|\Omega(t)\|_\infty \, dt = \infty.$$

Proof. If $\|\nabla u\|_\infty$ were bounded, (1) would bound the H^s norm up to T^*. Restarting the local solution at times approaching T^* gives a common positive extension length. The same energy estimate propagates every higher norm on a smaller common interval, so uniqueness gives an extension in the full smooth Sobolev class, contrary to maximality. A finite endpoint limsup would imply such a bound, since the solution is smooth on shorter intervals. The inequalities $\|A\|_{\mathrm{op}} \le \|A\|_{\mathrm{F}} \le \sqrt{d}\,\|A\|_{\mathrm{op}}$ identify the required Frobenius conclusion.

For the second assertion, incompressibility gives $\sum_i \partial_i \Omega_{ij} = \Delta u_j$. Consequently

$$\partial_k u_j = -\sum_i \partial_k \partial_i (-\Delta)^{-1} \Omega_{ij}.$$

Each second derivative of the Newton kernel is a principal-value kernel homogeneous of degree $-d$, smooth on the sphere and of spherical mean zero, together with a constant multiple of the delta distribution. For $0 < a < 1/2$, split its action into $|z| < a$, $a < |z| < 2$, and a smoothly cut off far part. In the first part subtract $\Omega(x)$. The $C^{0,\alpha}$ bound for Ω makes that part at most $C_d a^\alpha \|u\|_{H^s}$. The annular part and local delta term are at most $C_d\|\Omega\|_\infty(1+\log(2/a))$. In the far part use $\Omega_{ij} = \partial_i u_j - \partial_j u_i$ and integrate by parts. The differentiated cutoff kernel belongs to L^2, because it decays as $|z|^{-d-1}$; this part is bounded by $C_d\|u\|_2$. Taking $a = c(e+\|u\|_{H^s})^{-1/\alpha}$ gives

$$\|\nabla u\|_\infty \le C_d(1 + \|u(0)\|_2 + \|\Omega\|_\infty \log(e + \|u\|_{H^s})). \tag{2}$$

The finite sum over i, j, k is controlled by the once-counted norm $|\Omega|$ with a constant depending only on d. Set $Z = \log(e + \|u\|_{H^s})$. By (1) and (2), $Z' \leq C_d(1 + \|u(0)\|_2) + C_d\|\Omega\|_\infty Z$. A finite vorticity integral would bound Z by Gronwall, hence permit the same restart beyond T^*, a contradiction. For smooth fields the pointwise and essential suprema agree: a strict pointwise lower bound persists on an open ball of positive measure. Thus these are precisely the essential-supremum norms in H_d. $\square$

3 Full-dimensional solenoidal geometry

Let $E = \operatorname{span}(e_1, e_2, e_3)$ and let $\mathcal{R} = I_E \oplus (-I_{E^\perp})$. The fields used below have odd parity and satisfy $u(\mathcal{R}x) = \mathcal{R}u(x)$. They are fields on all of $\mathbb{R}^d$, not extensions constant in the extra coordinates.

Lemma 3.1 (Antisymmetric potentials)**.** *If $A_{ij} = -A_{ji}$ is a smooth compactly supported tensor and $v_j = \sum_i \partial_i A_{ij}$, then $\operatorname{div} v = 0$. For any trace-free matrix L,*

$$A_{ij}(x) = \frac{x_i(Lx)_j - x_j(Lx)_i}{d} \quad \Longrightarrow \quad \sum_i \partial_i A_{ij} = (Lx)_j.$$

Multiplying this tensor by an even compact Gevrey cutoff equal to one near zero produces an odd compact solenoidal velocity equal to Lx near zero. If $L\mathcal{R} = \mathcal{R}L$ and the cutoff is radial, the velocity is $\mathcal{R}$-equivariant.

Proof. Commutation of mixed derivatives and exchange of i, j make $\sum_{i,j} \partial_j\partial_i A_{ij}$ equal to its negative. For the displayed tensor, differentiation of the first numerator term gives $(d+1)(Lx)_j$, and differentiation of the second gives $(Lx)_j + x_j \operatorname{tr} L$. Their difference is $d(Lx)_j$. The cutoff preserves antisymmetry and compact support. The tensor is even, so its divergence is odd. Under $\mathcal{R}$ the tensor transforms as a two-tensor; applying divergence gives the stated equivariance. Gevrey bounds follow from the finite product rule with a polynomial. $\square$

The same replacement removes the use of curl from oscillatory solenoidalization. Put $D_i = \kappa\partial_{y_i} + m_i\partial_\theta$. These constant-coefficient operators commute. Thus $\sum_i D_i \sum_j D_j A_{ji} = 0$ for every antisymmetric A. When $m \cdot v = 0$, $|m| = 1$ and $F' = f$,

$$A_{ij} = (m_i v_j - m_j v_i)F(\theta) \quad \Longrightarrow \quad \sum_i D_i A_{ij} = v_j f(\theta) + \kappa \sum_i \partial_{y_i} A_{ij}.$$

The last term is the actual spatial corrector, retained also at the terminal truncation order. This identity treats every ambient component.

Lemma 3.2 (Symmetry of the ambient pressure and central blocks)**.** *If u is $\mathcal{R}$-equivariant, its Newton pressure obeys $P(\mathcal{R}x) = P(x)$. At $x \in E$, both $\nabla u(x)$ and $\nabla^2 P(x)$ preserve E and $E^\perp$.*

Proof. The chain rule and orthogonality show that $\sum_{i,j} \partial_i\partial_j(u_i u_j)$ is invariant under $\mathcal{R}$. The Newton kernel is radial and Lebesgue measure is orthogonally invariant, so a change of variables in its absolutely convergent convolution proves the pressure identity. Differentiate the velocity identity once and the pressure identity twice at a fixed point. The resulting maps commute with $\mathcal{R}$, and therefore preserve its two eigenspaces. No assertion that the active block is trace free follows or is used: the full trace alone vanishes. $\square$

Lemma 3.3 (The endpoint operator preserves the active plane)**.** *Along the central trajectory of an equivariant parent, suppose the normal $m(t)$ belongs to E. By Lemma 3.2, its gradient $M(t)$ and pressure Hessian $H(t)$ commute with $\mathcal{R}$. For the stationary displacement problem* *on* $[0,t_0]$ *with* $m(t)\cdot\eta(t)=0$, $\eta(0)=0$, *prescribed* $\eta(t_0)=Y$, *and coercive action* $\mathcal{A}(\eta)=\int_0^{t_0}(|\eta'|^2-\langle H\eta,\eta\rangle)dt$, *the endpoint map* $\Lambda Y=\operatorname{proj}_{m(t_0)^\perp}\eta'(t_0)$ *commutes with* $\mathcal{R}$. *In particular a prescribed endpoint in* $E\cap m(t_0)^\perp$ *produces a displacement and velocity in* E.

Proof. Reflection preserves the constraint, the two endpoints and the action. Uniqueness for the coercive variational problem therefore gives $\eta_{\mathcal{R}Y}=\mathcal{R}\eta_Y$. Taking the terminal derivative and its orthogonal projection proves the claim about Λ. If $Y=\mathcal{R}Y$, uniqueness also gives $\eta_Y=\mathcal{R}\eta_Y$ at every time. The identity remains true for the physical velocity $\eta'-M\eta$ because M commutes with the same reflection. Hence its extra components vanish exactly. This argument does not infer vanishing from an estimate that could be lost under the subsequent endpoint normalization. □

4 Polynomial bounds from the lifted particle flow

The estimates in this section control the particle flow through the small lifted transport field. The resulting bounds depend polynomially on the physical frequency.

Write $\mathcal{L}=\mathbb{R}^d\times\mathbb{R}$, with its product norm, and put $J_kx=(x,k\langle m_0,x\rangle)$, $|m_0|=1$. For a solved lifted packet $z(t,y,\theta)$ the lifted transport field is

$$b(t,y,\theta)=\big(\kappa z(t,y,\theta),\,\langle m_0,z(t,y,\theta)\rangle\big),\qquad k\kappa=1.$$

All occurrences of z in this section refer to the same finite packet plus its actual residual-removing correction. The normal component in the second entry retains the cancellation that gives the small drift estimate.

Lemma 4.1 (Actual graph invariance)**.** *Let* b *be a smooth bounded Lipschitz field on* $\mathcal{L}$, *continuous in time, of the displayed form. Its global flow* $\Psi_{s,t}$ *satisfies*

$$\Psi_{s,t}(J_kx)=J_k\psi_{s,t}(x),\qquad \psi_{s,t}(x)=\pi\Psi_{s,t}(J_kx),$$

where $\pi(y,\theta)=y$. *The map* ψ *is the actual flow of* $v(t,x)=\pi b(t,J_kx)$ *and* $\psi_{t,s}$ *is its inverse. For any* $\ell>0$, *the actual flow of* $v_\ell(t,x)=\ell v(t,x/\ell)$ *is* $\ell\psi_{s,t}(x/\ell)$.

Proof. The linear functional $q(y,\theta)=\theta-k\langle m_0,y\rangle$ satisfies $q(b)=0$. Its derivative along every trajectory is zero. Therefore the level set $q=0$, which is precisely the image of J_k, is invariant. Projection of the actual lifted ODE gives the ODE for v; uniqueness identifies its flow and inverse. Substitution and the chain rule prove the scaling identity. These arguments use no dimension-specific representation of a solenoidal field. □

Lemma 4.2 (Small lifted flow bounds)**.** *Suppose, for* $0\le t\le T$ *and every* $n\ge 0$,

$$\|D^nb(t)\|_\infty\le BR^n(n!)^2,\quad B\ge 0,\ R>0,\quad BRT\le\tfrac18.$$

Then the actual displacement $f_t=\Psi_{0,t}-\mathrm{id}$ *obeys*

$$\|D^nf_t\|_\infty\le Bt(4R)^n(n!)^2\quad(n\ge 0).\tag{3}$$

Put $\mathcal{R}(S)=(4R+1)((1+BT)S+2)$. *If also* $\|D^nb_t\|_\infty\le B_1R_1^n(n!)^2$ *with* $B_1\ge 0$ *and* $R_1\ge 0$, *then the actual material velocity and acceleration satisfy*

$$\|D^n(b(t)\circ\Psi_{0,t})\|_\infty\le B\mathcal{R}(R)^n(n!)^2,\tag{4}$$

$$\|D^n((b_t+Db\,b)(t)\circ\Psi_{0,t})\|_\infty\le(B_1+3B^2R)\mathcal{R}(4R+R_1)^n(n!)^2.\tag{5}$$

These estimates hold in every finite-dimensional normed space.

Proof. We use the finite generating-sum argument for the actual integral flow equation from [1, §3.8]. For a finite derivative cutoff N, let

$$F_N(t) = \sum_{n=1}^{N} \frac{r^n}{(n!)^2} \|D^n f_t\|_\infty, \qquad r = (4R)^{-1}.$$

The differentiated equation $f_t = \int_0^t b(s) \circ (\mathrm{id} + f_s)\, ds$ and the finite Faà di Bruno convolution give

$$F_N(t) \le \int_0^t \frac{BR(r + F_N(s))}{1 - R(r + F_N(s))}\, ds$$

as long as the denominator is positive. The algebra behind this inequality is the scalar factorial majorant for a composition: after division by $(n!)^2$, each partition coefficient is bounded by the corresponding geometric-series coefficient. It applies to operator norms of multilinear derivatives, independently of the ambient dimension. On the bootstrap region $F_N \le Bt$, one has $R(r + F_N) \le 1/4 + BRT \le 3/8$, so the integrand is at most $3B/5$. Continuity and a first-contact argument give $F_N(t) \le Bt$ for all $t \le T$, uniformly in N. The nth summand gives (3) for $n > 0$; order zero follows by directly integrating $\|b\|_\infty \le B$. Equivalently one may use the pointwise finite generating sums followed by their supremum; no differentiability of a supremum is required.

Adding the identity gives, for $n > 0$, $\|D^n \Psi_{0,t}\|_\infty \le (1 + BT)(4R + 1)^n (n!)^2$. The same finite composition formula implies that composing a field bounded by $CS^n(n!)^2$ with this map is bounded by $C\mathcal{R}(S)^n(n!)^2$. This gives (4). Differentiating a factorial bound once and using $(n+1)^2 \le 4^n$ bounds Db by $BR(4R)^n(n!)^2$. The factorial Leibniz convolution is at most three times the product of the two amplitudes. At the common radius $4R+R_1$ this bounds $b_t + Db\, b$ by $(B_1 + 3B^2R)(4R + R_1)^n(n!)^2$. The composition estimate proves (5). □

Lemma 4.3 (Physical labels and their inverse)**.** *The graph and physical scaling in Lemma 4.1 convert a bound $\|D^n f\|_\infty \le CS^n(n!)^2$ into*

$$\big\|D^n\big[\ell\pi f(J_k(\cdot/\ell))\big]\big\|_\infty \le C\big(\ell^{-1}S(1 + |k|)\big)^n (n!)^2 \quad (0 < \ell \le 1).$$

Consequently the physical displacement, material velocity and material acceleration have factorial bounds with amplitudes and radii polynomial in $1+B+R+B_1+R_1+T+\ell^{-1}+|k|$. If the physical field is solenoidal, its inverse deformation, evaluated in particle labels, has factorial bounds polynomial in the same parameters, with exponents allowed to depend on d.

Proof. The chain rule for a linear inner map gives the factor $\|J_k/\ell\|^n$, and $\|J_k\| \le 1 + |k|$. The outer map has norm at most $\ell \le 1$. This proves the first assertion, including order zero. Apply it to (3)–(5). The first spatial derivative of a flow is the identity plus the derivative of its displacement, so it has the same type of polynomial factorial bound after enlarging the parameters.

Solenoidality gives $\det D\psi = 1$ by the variational equation and Jacobi's formula. Thus $(D\psi)^{-1} = \mathrm{adj}(D\psi)$. Every entry of the adjugate is a sum of $(d-1)!$ products of $d-1$ entries. For a product of r factorially bounded fields, the multinomial Leibniz formula and $\prod_{i=1}^r n_i! \le n!$ show that its nth derivative is bounded by $C_d C^r (rR)^n (n!)^2$. Here $r = d - 1$ is fixed. Hence the inverse deformation has a polynomial factorial bound, with no exponential in the physical frequency. The same reasoning applies to composition with the given parent flow: the finite composition and product formulas involve its existing factorial bounds and the actual graph-flow bounds just proved. Time differentiation twice uses the actual material velocity and acceleration above. □

The trace identity needed in this construction is also dimension independent. If a lifted field is tangent to the graph, $b = J_k \pi b$. Differentiating and cyclically commuting the trace gives

$$\mathrm{tr}_{\mathbb{R}^d}(\pi Db\, J_k) = \mathrm{tr}_{\mathcal{L}}(J_k \pi Db) = \mathrm{tr}_{\mathcal{L}}(Db).$$

Thus lifted divergence zero implies ordinary divergence zero for the graph velocity. Scaling preserves it. When the graph flow is joined to the parent particle map, both maps preserve ordinary d-dimensional volume, and their composition does as well.

4.1 Cylinder and graph bounds in L^2

The preceding supremum estimates also give the needed quantitative L^2 estimates; a supremum bound alone would not do so on the whole space. Let b be periodic in its last variable with period P, and assume its actual cylinder divergence is zero. Suppose the input bounds of Lemma 4.2 hold both in supremum norm and in $L^2(\mathbb{R}^d \times \mathbb{T}_P)$, with common amplitudes B, B_1 and radii R, R_1. These input L^2 bounds are supplied by the packet's weighted Sobolev estimates, including the true time derivative.

The flow satisfies $\Psi_t(y, \theta + P) = \Psi_t(y, \theta) + (0, P)$ by uniqueness. It therefore descends to the cylinder. Its variational Jacobian obeys

$$\partial_t \det D\Psi_t = \big(\operatorname{div} b\big)(t, \Psi_t) \det D\Psi_t = 0.$$

The inverse flow exists by the same bounded Lipschitz ODE, so change of variables gives

$$\|a \circ \Psi_t\|_{L^2(\mathbb{R}^d \times \mathbb{T}_P)} = \|a\|_2. \tag{6}$$

For the lifted velocity of the actual corrected packet, $\operatorname{div} b = \kappa \operatorname{div}_y z + \partial_\theta(m_0 \cdot z) = 0$ is exactly its already proved closed solenoidal constraint, promoted to a pointwise identity by smoothness.

Set $S_v = \mathcal{R}(R)$ and $S_a = \mathcal{R}(4R + R_1)$, with $\mathcal{R}$ as in Lemma 4.2. Then the actual fields satisfy

$$\|D^n(b(t) \circ \Psi_t)\|_2 \le B S_v^n (n!)^2, \tag{7}$$

$$\|D^n((b_t + Db\, b)(t) \circ \Psi_t)\|_2 \le (B_1 + 3B^2 R) S_a^n (n!)^2, \tag{8}$$

$$\|D^n(\Psi_t - \mathrm{id})\|_2 \le Bt S_v^n (n!)^2. \tag{9}$$

To prove these assertions, expand the derivative of a composition into its finite Faà di Bruno sum. In each term place the single factor $D^r a \circ \Psi_t$ in L^2 using (6); place all derivatives of the inner flow in L^∞ using Lemma 4.2. The numerical composition convolution is then exactly the one used for the supremum bound, so its radius is $\mathcal{R}(S)$ for an outer radius S. For $Db\, b$, put one factor in L^2 and the other in L^∞ in each Leibniz term. The factorial convolution costs at most three, giving amplitude $3B^2 R$ at radius $4R$. This proves the first two estimates. Finally integrate the actual identity $\Psi_t - \mathrm{id} = \int_0^t b(s) \circ \Psi_s\, ds$ and apply Minkowski's inequality to each derivative. This proves the last one, including order zero. In particular its stated radius is S_v; no claim that the L^2 displacement has radius $4R$ is needed.

Here is the graph trace with the scaling factors retained. If a smooth periodic field a has $\|D^n a\|_2 \le C S^n (n!)^2$, then

$$\|D^n\{\ell \pi a(J_k(\cdot/\ell))\}\|_{L^2(\mathbb{R}^d)} \le C_P \ell^{1+d/2} C(1+S) \left(\frac{4S(1+|k|)}{\ell}\right)^n (n!)^2. \tag{10}$$

Indeed the one-dimensional periodic estimate

$$\sup_\theta |v(\theta)|^2 \le \frac{2}{P} \int_{\mathbb{T}_P} |v|^2 + 2P \int_{\mathbb{T}_P} |v_\theta|^2$$

follows from the fundamental theorem of calculus, the triangle inequality and Cauchy–Schwarz. The same constants apply to any finite-dimensional normed target, in particular to the operator norm of $D^n a$, independently of n. Applied at each spatial point it bounds evaluation on any phase graph by $C_P(\|a\|_2 + \|\partial_\theta a\|_2)$; the phase slope does not enter this trace estimate. Differentiating the affine graph costs $\|J_k\|^n \le (1+|k|)^n$ before applying that trace. Its one extra angular derivative is bounded using $(n+1)^2 \le 4^n$. Finally $x = \ell y$ gives the exact factor $\ell^{1+d/2-n}$. For $0 < \ell \le 1$ the prefactor $\ell^{1+d/2}$ may be discarded upwards. This proves (10). The argument applies separately to the actual displacement, material velocity and material acceleration in (7)–(9).

For the bounds used in the parameter induction, take $0 < \chi < 1/100$ and suppose the source bounds satisfy $T, R, R_1, B_1 \le k^\chi$, while $B \le 2k^{-1/2}$ and $BRT \le 1/8$. The two composition radii satisfy $S_v, S_a \le Ck^{2\chi}$. Thus the graph amplitudes for displacement and velocity are at most $Ck^{-1/2+3\chi}$ and $Ck^{-1/2+2\chi}$, respectively; the acceleration amplitude is at most $Ck^{3\chi}$. Their graph radii are at most $C\ell^{-1}k^{1+2\chi}$. After one fixed numerical lower bound on k, these give the convenient bounds

$$\|D^n a\|_{2,\infty},\ \|D^n b_{\mathrm{mat}}\|_{2,\infty} \le k^{-1/4}(\ell^{-1}k^{5/4})^n (n!)^2, \qquad \|D^n c_{\mathrm{mat}}\|_{2,\infty} \le k^{1/4}(\ell^{-1}k^{5/4})^n (n!)^2.$$

Here $a, b_{\mathrm{mat}}, c_{\mathrm{mat}}$ are the actual physically scaled relative displacement, material velocity and material acceleration. These three fields are obtained from the same lifted flow and graph restriction.

4.2 Fixed Sobolev shifts

The passage from L^2 jets to supremum jets uses only a fixed number of additional derivatives. Let $q > d/2$ be an integer, let $r \ge 0$ be fixed, and suppose a smooth finite-dimensional vector-valued field on $\mathbb{R}^d$ satisfies

$$\|D^n a\|_2 \le AR^n (n!)^2 \qquad (n \ge 0).$$

Then

$$\|D^n D^r a\|_\infty \le C_{d,q,r} A (1+R)^{q+r} (4R)^n (n!)^2 \qquad (n \ge 0). \tag{11}$$

Indeed, apply the H^q embedding to $D^{n+r}a[v_1, \dots, v_{n+r}]$ for arbitrary fixed unit vectors v_i, and then take their supremum. Each of its derivatives of order $j \le q$ is bounded in L^2 by $\|D^{n+r+j}a\|_2$. The embedding constant is independent of n and of the vectors v_i. Finally

$$((n+r+j)!)^2 \le 4^{n+r+j}(n!)^2((r+j)!)^2$$

gives (11) after summing the fixed range $0 \le j \le q$.

In particular, if the composed particle displacement, velocity and acceleration have L^2 Gevrey amplitude at most k and radius at most k^2, then their first spatial derivatives have supremum amplitude at most k^{2q+4} and radius at most $4k^2$, for a sufficiently large fixed lower bound on k. The additive identity in the deformation matrix is absorbed by the same bound. This applies to the second frame coefficient $F_2 = DW_0$, where W_0 is the material acceleration. The shift depends only on the fixed dimension and derivative order; it is independent of the packet grade and truncation.

5 The active three-plane and ambient amplification

All cross products in this section are taken solely in the oriented Euclidean three-plane E. They define an orthonormal basis of E; they are never used as a divergence or pressure identity on $\mathbb{R}^d$. Write $M = \nabla u(t, 0)$ for the parent central gradient and $H = \nabla^2 P(t, 0)$.

Oddness fixes the central particle at zero. Reflection makes M, H block diagonal on $E \oplus E^\perp$. The active block need not be trace free.

The geometric input at a stage is the decomposition

$$M = B + hqp^T + \mathcal{E}, \qquad |\mathcal{E}| \le E_*, \qquad h(t_0) = h_*, \tag{12}$$

where p, q are the normalized transported normal and primary velocity of the preceding packet. They lie in E and are orthogonal; put $n = p \times q$. The shear is the actual preceding packet amplitude, $h = c_0|m_{\rm old}||v_{\rm old}|$ with constant $c_0 > 0$. Consequently $\dot h/h = -B_{pp} - B_{qq}$, by the two transport equations for the preceding normal and velocity. Here B is the gradient of the preceding background solution. On the stage interval $|B| \le CG_*$ and $|\dot B| \le CG_*^2$. The transported normal and velocity satisfy

$$\dot m = -M^T m, \qquad \dot v = -Mv + 2m\frac{m \cdot Mv}{|m|^2}, \qquad m \cdot v = 0. \tag{13}$$

The coefficient 2 follows by differentiating the last constraint. These are full ambient equations; for central data in E they preserve E.

At activation put

$$a = p \cdot Bq \in [1/2, 2], \quad \beta = (n \cdot Bq)/a, \quad \tfrac{1}{2} \le \beta x_{\rm prev}^2 \le 2,$$
$$\epsilon = \sqrt{a/h_*}, \quad \tau = a(t - t_0)/\epsilon, \quad x = \sqrt{\beta}\,\tau.$$

The target is $x = x_{\rm tar} \ge 2$. Let $\Theta \ge 2 + \beta^{-1/2} + x_{\rm tar}/\sqrt{\beta}$ bound the scaled time, with at most Θ^{-60} scaled time retained after the target. The geometric error parameter is

$$e = C_d\big(\epsilon\Theta G_*^2 + E_* + \rho K_h^{c_d}/\epsilon\big), \qquad e\Theta^{60} \le c_d x_{\rm prev}^{-10}. \tag{14}$$

Here ρ is the physical packet length and K_h bounds the actual parent particle map, its inverse deformation and required derivatives. The small constant on the right is fixed before the scales are chosen.

Lemma 5.1 (Endpoint selection)**.** *On a positive history interval $[0, t_0]$, assume the one-sided pressure bound $H \le K_+ I$ and $K_+ t_0^2/2 \le 1/2$, provided by the mean reserve. Suppose $\|M\|_\infty \le C_M h_*$, $\|H\|_\infty \le C_H h_* h_{\rm old}$, $1 \le h_{\rm old} \le h_*$, and $t_0^{-1} \le h_*$. Suppose also $\overline{B} = B(t_0) + \mathcal{E}(t_0)$ has $\overline{B}_{pp} < 0$ and $\|\overline{B}\| \le \zeta h_*$. For sufficiently small ζ, depending only on C_M, C_H, an endpoint $Y \in E \cap n^\perp$ of size at most C/h_* gives a stationary displacement whose terminal primary velocity satisfies*

$$v_q = 1, \qquad -C \le v_p \le 0, \qquad v_{E^\perp} = 0.$$

The displacement is stationary in the full transverse space.

Proof. Choose $m_0 = F(t_0, 0)^T n/|F(t_0, 0)^T n|$ and transport it by F^{-T}. Let Λ be the endpoint operator of Lemma 3.3. Integration by parts in its stationary equation gives

$$Y \cdot \Lambda Z = \int_0^{t_0} (\eta_Y' \cdot \eta_Z' - H\eta_Y \cdot \eta_Z)\, dt.$$

The stationary paths start at zero. The inequality $\int |\eta|^2 \le (t_0^2/2) \int |\eta'|^2$ therefore makes their action nonnegative even for a nonzero terminal value. Thus Λ is symmetric and positive semidefinite. To bound it, put $L_0 = t_0^{-1} + \|M\|_\infty + \|H\|_\infty^{1/2} + 1$. A comparison path vanishes before $t_0 - L_0^{-1}$ and then equals $L_0(t - t_0 + L_0^{-1})\,\mathrm{proj}_{m(t)^\perp} Y$. Since $\|\partial_t \mathrm{proj}_{m^\perp}\| \le C\|M\|$, its action is at most $C(L_0 + \|M\|_\infty^2/L_0 + \|H\|_\infty/L_0)|Y|^2 \le C_0 h_* |Y|^2$. Minimality yields $0 \le \Lambda \le C_0 h_* I$ on the whole transverse space, and reflection reduces its restriction to $E \cap n^\perp$ to a symmetric two-by-two matrix.

Write $b=\Lambda_{pq}$, $d_0=\Lambda_{qq}$ and $u_*=\Lambda_{pp}-\overline{B}_{pp}>0$. The terminal velocity $v=\eta'-M\eta$ has active coordinates

$$\begin{pmatrix} v_p \\ v_q \end{pmatrix} = \begin{pmatrix} u_* & b-\overline{B}_{pq} \\ b-h_*-\overline{B}_{qp} & d_0-\overline{B}_{qq} \end{pmatrix} \begin{pmatrix} Y_p \\ Y_q \end{pmatrix}.$$

If $b\le h_*/2$, take $Y=-p$. Then $v_p=-u_*\le 0$ and $v_q=h_*+\overline{B}_{qp}-b\ge h_*/3$. If $b>h_*/2$, positivity implies $\Lambda_{pp}\ge b^2/d_0\ge h_*/(4C_0)$. Take $Y=q-(b-\overline{B}_{pq})p/u_*$. Its norm is bounded by a constant depending on C_0, and $v_p=0$. Moreover

$$v_q=\left(d_0-\frac{b^2}{u_*}\right)-\overline{B}_{qq}+\frac{bh_*+b(\overline{B}_{pq}+\overline{B}_{qp})-h_*\overline{B}_{pq}-\overline{B}_{qp}\overline{B}_{pq}}{u_*}.$$

The parenthesis is nonnegative because $u_*\ge\Lambda_{pp}$. The numerator is at least $h_*^2/2-C(C_0+1)\zeta h_*^2$. Since $u_*\le(C_0+\zeta)h_*$, choosing ζ small gives $v_q\ge ch_*$. Divide the displacement, endpoint and velocity by this positive terminal value. Both cases give the asserted normalization and endpoint bound. Reflection gives exactly zero extra components before and after normalization. At $t_0=0$ the prescription is simply $v=q$, and no stationary history problem is needed. ☐

Lemma 5.2 (The scalar growing solution)**.** *Let $0<\beta\ll 1$, $P_0=\beta\tau^2$, $Q_0=-2\beta\tau$, and $D=1+P_0^2$. The equation*

$$(DV')'=2(1-\beta P_0)V, \qquad V_\lambda(0)=1, \qquad V_\lambda'(0)=\lambda\ge 0$$

has a positive solution to every target under consideration, and $V_\lambda(x=1)\ge e^{b/\sqrt{\beta}}$. For $x\ge 1$ put $r_\lambda=-V_\lambda'/V_\lambda$. There is a bounded nonnegative z such that

$$r_\lambda=\frac{\sqrt{\beta}}{x}-\frac{z}{x^2}, \qquad z^2=\frac{2}{1+x^{-4}}+O(\sqrt{\beta}) \quad (x\ge 2).$$

Writing $g(\tau)=V_0(\tau)$, the two-component propagator for $(-V',V)$ obeys

$$\|\Phi_0(t,s)\|\le C\Theta^8 g(t)/g(s).$$

All statements concern a scalar equation and are dimension independent.

Proof. On $x\le 1$, integrating twice with $D\le 2$ and $2(1-\beta P_0)\ge 1$ gives $V_\lambda\ge 1+\lambda\tau/2+\frac{1}{2}\int_0^\tau(\tau-s)V_\lambda(s)ds$. Positive iteration bounds it below by $\cosh(\tau/\sqrt{2})$. Its logarithmic derivative l satisfies $0\le l$ and $l'\le 2-l^2$, whence $l\le\sqrt{2}\coth(\sqrt{2}\tau)$. For $x\ge 1$, first write $\widehat{V}(x)=V(x/\sqrt{\beta})$, and then put $s=1-\rho$, $\rho=x^{-1}$ and $f(\rho)=\widehat{V}(1/\rho)/\rho$. Direct substitution gives

$$((1+\rho^4)f_\rho)_\rho=(2/\beta-2\rho^2)f.$$

Starting from $f(1)>0$, $f_\rho(1)<0$, backward integration preserves both signs. For $z=-\sqrt{\beta}f_\rho/f$ and $\mu=\sqrt{(2-2\beta\rho^2)/(1+\rho^4)}$,

$$\sqrt{\beta}z_s=\mu^2-z^2+\sqrt{\beta}\frac{4\rho^3}{1+\rho^4}z.$$

This gives a uniform upper bound and preserves nonnegativity. Indeed its initial value at $\rho=1$ is $z(1)=\sqrt{\beta}+V_\lambda'(1/\sqrt{\beta})/V_\lambda(1/\sqrt{\beta})$, which is bounded independently of λ by the preceding logarithmic-derivative estimate. The equation for $z-\mu$ has damping coefficient $z+\mu\ge c>0$ and a bounded forcing multiplied by $\sqrt{\beta}$. Its integrating factor gives $|z-\mu|\le Ce^{-c(1-\rho)/\sqrt{\beta}}+C\sqrt{\beta}$, which proves the direction formula for $\rho\le 1/2$.

Reduction of order gives $V_\lambda = g(1+\lambda I)$, where $I(\tau) = \int_0^\tau (Dg^2)^{-1}$ and $I(1) \geq c > 0$. Positivity before $x = 1$, and increase of xg afterwards, give $g(s)/g(u) \leq C\Theta$ for $s \leq u$. For arbitrary scalar data at s,

$$V(t) = \frac{g(t)}{g(s)}\left[V(s) + D(s)\left(V'(s) - \frac{g'(s)}{g(s)}V(s)\right)\int_s^t \frac{(g(s)/g(u))^2}{D(u)}\,du\right].$$

The integral is at most $C\Theta^3$, $D(s) \leq C\Theta^4$, and $|g'/g| \leq C$. Near $\tau = 0$ the latter bound follows from $g'(0) = 0$ and $l' \leq 2 - l^2$; away from zero it follows from the displayed bounds for l and z. Differentiating this formula costs at most one further power of Θ, proving the propagator assertion. This is the scalar calculation in [1, §§4.4–4.5], reproduced to specify exactly the part of that argument used here. □

The selected initial conditions at activation are $m(t_0) = s_0 n(t_0)$, $v_q(t_0) = 1$, $v_n(t_0) = 0$ and $-C \leq v_p(t_0) \leq 0$. At the central label express $m = s_0(Pp+\epsilon Qq+Nn)$ and $v = \epsilon Up+Vq+\epsilon Wn$. The moving-frame connection has $(S_f)_{12} = B_{pq}$, $(S_f)_{13} = B_{pn}$ and $(S_f)_{32} = -B_{nq}$; skew symmetry determines its other entries. Equations (13) then give

$$(P,Q,N) = (\beta\tau^2, -2\beta\tau, 1) + O(e\Theta^5), \qquad W = -(PU+QV)/N, \quad N \geq 1/2.$$

The reference ray system is $P' = -Q$, $Q' = -2\beta N$, $N' = 0$. Its triangular propagator costs $C\Theta^2$. The exact error integral is at most $Ce\Theta^5 + Ce\Theta^3 \sup|\text{error}|$, and the last term is absorbed by (14). Substitution in the velocity equation gives a coefficient error at most $Ce\Theta^{13}$ from the two-component ideal system

$$U' = -2V + \frac{2P_0((P_0+\beta)V + Q_0U)}{1+P_0^2}, \qquad V' = -U.$$

Indeed the only unsuppressed contributions to $J = (m \cdot Mv)/(s_0 a)$ are $PB_{pq}V/a$, $h\epsilon^2 QU/a$ and $NB_{nq}V/a$. The identity $\dot{h}/h = -B_{pp} - B_{qq}$ gives $h/h_* = 1 + O(\epsilon\Theta G_*)$ on this interval, so $h\epsilon^2/a = 1 + O(\epsilon\Theta G_*)$. The remaining contributions contain either ϵG_* or the error in (14). Transversality bounds $|W| \leq C\Theta^2(|U|+|V|)$; the denominator $|m|^2/s_0^2$ is at least $1/4$. These observations bound every rational coefficient without using a trace condition on the active block. The full matrix estimate, including all transverse coordinates, is derived in Section 8.

Weighting Duhamel by $g(s)/g(t)$, Lemma 5.2 makes its error operator at most $Ce\Theta^{22}$. A geometric series therefore gives the exact active propagator bound $2C\Theta^8 g(t)/g(s)$. For the selected data, $\lambda = -v_p(t_0)/\epsilon \geq 0$,

$$|U - U_\lambda| + |V - V_\lambda| \leq Ce\Theta^{30}(1+\lambda)g.$$

For $\tau \geq 1$, $V_\lambda \geq c(1+\lambda)g$, so this is a relative error. In particular

$$J/V = P_0 + \beta + Q_0 r_\lambda + O(e\Theta^{34}) > 0. \tag{15}$$

For $x \leq 1$ its leading term is at least β, since $Q_0, r_\lambda \leq 0$. For $x \geq 1$ it equals $x^2 - \beta + 2\sqrt{\beta}z/x$. The error is less than half the lower bound by (14).

Lemma 5.3 (Central frame renewal)**.** *At the target set $p_2 = m/|m|$, $q_2 = v/|v|$, $n_2 = p_2 \times q_2$, $a_2 = p_2 \cdot Mq_2$ and $\beta_2 = (n_2 \cdot Mq_2)/a_2$. Then*

$$\begin{aligned} a_2/a &= 1 + O(x_{\text{tar}}^{-4} + \beta x_{\text{tar}}^{-2} + \sqrt{\beta}x_{\text{tar}}^{-3} + e\Theta^{40}),\\ \beta_2 x_{\text{tar}}^2 &= 1 + O(\sqrt{\beta} + x_{\text{tar}}^{-4} + e\Theta^{40}),\\ p_2 \cdot Mp_2 &\leq -\frac{c\sqrt{h_*}}{x_{\text{prev}}x_{\text{tar}}} + C(G_* + E_*). \end{aligned}$$

Moreover $A_{\text{tar}} = |m||v|$ at the target satisfies $A_{\text{tar}} \geq K_h^{-c}\Theta^{-c}e^{bx_{\text{prev}}}$.

Proof. Put $r = U/V$, $w = W/V$, $D_a = |m|^2/s_0^2$ and $E_v = 1 + \epsilon^2(r^2 + w^2)$. Normalize the two vectors directly:

$$a_2/a = \frac{J/V}{\sqrt{D_a}\sqrt{E_v}}, \quad \beta_2 = \frac{[(P, \epsilon Q, N) \times (\epsilon r, 1, \epsilon w)] \cdot (M(v/V)/a)}{(J/V)\sqrt{E_v}}.$$

In the second numerator substitute $P = x^2$, $Q = -2\sqrt{\beta}x$, $r = \sqrt{\beta}/x - z/x^2$. The terms of size βx^2 and $\sqrt{\beta}xz$ cancel, leaving

$$-1 + (1 + x^{-4})z^2 + \beta x^{-2} - 2\sqrt{\beta}zx^{-3} = 1 + O(\sqrt{\beta}).$$

The perturbation contributes at most $Ce\Theta^{36}$. Also $J/V = x^2 - \beta + 2\sqrt{\beta}z/x + O(e\Theta^{34})$, $D_a = 1 + x^4 + O(e\Theta^7)$ and $E_v = 1 + O(\epsilon^2\Theta^4)$. These identities prove the first two estimates. The rank-one part of (12) gives $p_2 \cdot Mp_2 = h\epsilon QP/D_a + O(G_* + E_*)$. Since $QP \le -c\sqrt{\beta}x^3$, $D_a \le Cx^4$ and $h\epsilon \asymp \sqrt{h_*}$, this proves compression. Finally $V_\lambda(x_{\text{tar}}/\sqrt{\beta}) \ge x_{\text{tar}}^{-1}V_\lambda(1/\sqrt{\beta}) \ge x_{\text{tar}}^{-1}e^{b/\sqrt{\beta}}$ and $s_0^{-1} \le K_h^c$; the relative-error estimates give the lower bound for A_{tar}. ☐

Let $\mathcal{P} \ge 1$ bound the source quantities $K_h, \Theta, \epsilon^{-1}, h_{\text{child}}, \delta^{-1}$ and the fixed coefficient and inverse costs. These quantities are estimated in Section 8 before the frequency is selected. The choice $\alpha = \delta h_{\text{child}}/A_{\text{tar}}$ therefore satisfies $\alpha \le \mathcal{P}^c e^{-bx_{\text{prev}}}$. At the central target the cutoff equals one, the phase is zero and $f_\delta'(0) = \delta^{-1}$, so the leading gradient is exactly $\alpha\delta^{-1}v \otimes m = h_{\text{child}}q_2p_2^T$. Section 8 proves the corresponding full transverse propagator and pressure-sign estimates in a neighborhood of the central label. The following consequence of that comparison gives a bound uniform over the stages of the iteration.

Lemma 5.4 (Uniform target-size comparison)**.** *For the selected primary, on every label in the quantitative packet neighborhood, the full ambient relative estimates of Section 8 give*

$$\frac{|m(t, y)||v(t, y)|}{A_{\text{tar}}} \le C_d \qquad (1 \le \tau \le \tau_{\text{end}}).$$

On the history interval and on $0 \le \tau < 1$ the same ratio is at most $K_h^{c_d}\Theta^{c_d}\epsilon^{-c_d}e^{-bx_{\text{prev}}}$. For $g(\tau) = V_0(\tau)$ and $\alpha = \delta h_{\text{child}}/A_{\text{tar}}$, one also has $\sup \alpha g \le \mathcal{P}^{c_d}$ and $\alpha \le \mathcal{P}^{c_d}e^{-bx_{\text{prev}}}$.

Proof. The full relative comparison, including its extra components, and $\epsilon\Theta^2 \ll 1$ imply, for $\tau \ge 1$,

$$|m(t, y)||v(t, y)| \asymp_d s_0\sqrt{1 + x^4}\, V_\lambda(x/\sqrt{\beta}).$$

The small perturbations of the initial label and of s_0 are included in the same constants. At the central target this equivalence bounds A_{tar} from below by a fixed multiple of the right side. These constants do not grow with Θ or the stage.

For $0 \le x \le 1$, V_λ is increasing. For $x \ge 1$ the reciprocal-variable proof of Lemma 5.2 shows that $xV_\lambda(x/\sqrt{\beta})$ is increasing. Hence for $1 \le x \le x_{\text{tar}}$,

$$(1 + x^2)V_\lambda(x/\sqrt{\beta}) \le (x + x^{-1})x_{\text{tar}}V_\lambda(x_{\text{tar}}/\sqrt{\beta}) \le 2x_{\text{tar}}^2V_\lambda(x_{\text{tar}}/\sqrt{\beta}).$$

For $x \le 1$ the same conclusion follows by comparison first with $V_\lambda(1/\sqrt{\beta})$ and then with the target. Since $\sqrt{1 + x^4} \asymp 1 + x^2$, this proves the constant size comparison up to the target. Afterwards the logarithmic derivative in τ of $\sqrt{1 + x^4}V_\lambda$ is

$$\frac{2\sqrt{\beta}x^3}{1 + x^4} - \frac{\sqrt{\beta}}{x} + \frac{z}{x^2},$$

which is bounded for $x \ge x_{\text{tar}} \ge 2$. The retained scaled interval has length at most Θ^{-60}, so its extra factor is at most two. The full relative errors preserve this fixed bound.

The stationary history inverse bounds the unnormalized history by $K_h^{c_d}$. On $\tau \le 1$, the scaled system has bounded coefficients and initial norm at most $C(1+\lambda)$, with $\lambda \le C/\epsilon$; converting the coordinates therefore costs only the displayed old-data polynomial. Divide these estimates by the exponential target lower bound from Lemma 5.3 to obtain the early and history ratio. Finally $V_0 \le V_\lambda$. The monotonicity argument just used gives $\sup g \le C x_{\mathrm{tar}} V_\lambda(x_{\mathrm{tar}}/\sqrt{\beta})$, whereas $A_{\mathrm{tar}} \ge c_d s_0 x_{\mathrm{tar}}^2 V_\lambda(x_{\mathrm{tar}}/\sqrt{\beta})$. Their growing factors cancel in αg. The remaining factors $\delta, h_{\mathrm{child}}, s_0^{-1}, x_{\mathrm{tar}}$ have the fixed polynomial source bounds. The target exponential lower bound similarly proves the assertion for α itself. $\square$

6 The finite packet and its exact Euler correction

This section concerns one prescribed smooth parent solution on a closed interval $[0,T]$. All constants are allowed to depend on this parent, on d, and on the prescribed seed. The higher high coefficients use the variational history and forward-tail construction of [1, §3]. All source constants are fixed before the packet frequency is chosen. A bound on these constants along an infinite sequence of parents is a separate question; none is asserted in this section.

Let U be the parent, let P_U be its Newton pressure, and let Φ_t be its volume-preserving flow, with $\Phi_0 = \mathrm{id}$. In particle labels put

$$F = D\Phi_t, \qquad M = (DU)\circ\Phi_t = F_t F^{-1}, \qquad H = (D^2 P_U)\circ\Phi_t = -F_{tt}F^{-1}. \tag{16}$$

We assume that these are the actual flow and coefficient fields, with bounded smooth spatial derivatives and continuous coefficient paths on $[0,T]$. For the quantitative assertions we assume a Gevrey bound on the known coefficients F, F^{-1}, F_t, F_{tt}:

$$\sup_{t,y} \|D_y^j C(t,y)\| \le C_C R_C^j (j!)^2, \qquad C \in \{F, F^{-1}, F_t, F_{tt}\}, \quad j \ge 0. \tag{17}$$

These are conditions on the current known parent, not on the child being constructed. The positive lower bound of F follows from the actual inverse. All derivative norms in this section are operator norms unless specified otherwise.

Fix a unit vector $m_0 \in \mathbb{R}^d$ and set

$$m(t,y) = F(t,y)^{-\top} m_0.$$

Then $m \ne 0$ and $m_t = -M^\top m$. The phase variable θ belongs to $\mathbb{T}_P = \mathbb{R}/P\mathbb{Z}$, with $P = 2\pi$ when a specific periodic seed is used. Every vector field below has all d components. The orthogonal space $m^\perp$ has dimension $d-1$; it is never replaced by a chosen one- or two-dimensional subspace.

For physical Sobolev persistence one may use $s_d = \lfloor d/2 \rfloor + 3 > d/2+1$, as in the preceding section. Packet estimates also involve the cylinder $\mathbb{R}^d \times \mathbb{T}_P$. To avoid any implicit three-dimensional threshold we use the larger fixed base order

$$b_d = 2(d+2)+1 = 2d+5. \tag{18}$$

The cylinder embedding, product, and commutator estimates at this order follow by Fourier series in θ and the ordinary Sobolev estimates in y. All their constants depend on d and P, not on the number of additional derivatives.

6.1 The mean inverse

Let $\mathcal{H} = L^2_\sigma(\mathbb{R}^d;\mathbb{R}^d)$ be the closed solenoidal subspace. Choose a nonnegative even smooth cutoff χ, equal to one near zero and supported in the unit ball, and put $\chi_\ell(y) = \chi(\ell y)$. There is a bounded map from the homogeneous first-order tensor space to L^2 defined first on compact smooth tensors by

$$(T_{\chi_\ell}Q)_j = \sum_{i=1}^{d} \partial_i\{\chi_\ell(Q_{ij} - Q_{ji})\}, \qquad A_{\chi_\ell} = T_{\chi_\ell}T^*_{\chi_\ell}.$$

The tensor space here is the completion for the square sum of all $\partial_k Q_{ij}$, with $1 \le i, j, k \le d$. Hardy's inequality for $d \ge 3$ controls the term $(\partial_i\chi_\ell)Q_{ij}$ and extends T continuously. Its bound is uniform for $0 < \ell \le 1$ after dilation. Consequently A_{χ_ℓ} is positive and symmetric, its range is solenoidal, and every output vanishes off $\operatorname{supp}\chi_\ell$. This is a bounded operator with spatially supported outputs; compactness as an operator on L^2 is neither needed nor asserted.

We record the localization inequality used below. There are constants $C_{1,d}, C_{2,d}$ and a fixed $a_d > 0$ such that, if

$$\begin{aligned}
&\langle M(0, y)z, z\rangle \ge -B_e|z|^2 && \text{for } |y| \ge a_d r/\ell,\\
&\langle M(0, y)z, z\rangle \ge -B_c|z|^2 && \text{for } |y| < a_d r/\ell,
\end{aligned}$$

where $0 \le r \le 1/4$, then, for $z \in \mathcal{H}$ and $\mathcal{L} \ge C_{1,d}B_c$,

$$\langle M(0)z, z\rangle_{L^2} + \mathcal{L}\langle A_{\chi_\ell}z, z\rangle_{L^2} \ge -(B_e + C_{2,d}B_c r^d)\|z\|_2^2. \tag{19}$$

Here and below constants absorb the fixed normalization of the cutoff plateau. To obtain (19), set $Q = T^*_{\chi_\ell}z$. On the plateau, subtract one half of the uncut antisymmetric divergence of Q from z. The difference is weakly harmonic, by the defining Riesz identity for Q. Its L^2 mass on the concentric ball of relative radius r is bounded by $C_d r^d$ times its mass on the plateau ball. This follows from the interior harmonic estimate, obtained by nested cutoff energy estimates and Sobolev embedding, and then by dilation. The other part is controlled by $\|Q\|^2_{\dot H^1} = \langle A_{\chi_\ell}z, z\rangle$. Combining this local bound with the exterior lower bound proves the inequality. The argument uses the complete tensor, so no special three-dimensional curl formula occurs.

Assume $H(t, y) \le KI$ in quadratic-form order and

$$K\frac{T^2}{2} + (B_e + C_{2,d}B_c r^d)T \le \frac{1}{2}. \tag{20}$$

For $v \in L^2(0, T; \mathcal{H})$ define

$$Iv(t) = -\int_t^T v(s)\,ds, \qquad X_v = FIv, \qquad D_F v = \partial_t X_v = F_t Iv + Fv.$$

The mean form is

$$\begin{aligned}
\dashv(v, w) = \int_0^T \langle D_F v, D_F w\rangle\,dt - \int_0^T \langle HX_v, X_w\rangle\,dt&\\
+ \langle (M(0) + \mathcal{L}A_{\chi_\ell})X_v(0), X_w(0)\rangle.&
\end{aligned} \tag{21}$$

Spatial integrations are included in each inner product. The form need not be symmetric because $M(0)$ need not be symmetric. It is bounded and coercive. Indeed $X_v(T) = 0$, hence

$$\|X_v\|^2_{L^2_t L^2_y} \le \frac{T^2}{2}\|D_F v\|^2_{L^2_t L^2_y}, \qquad \|X_v(0)\|_2^2 \le T\|D_F v\|^2_{L^2_t L^2_y}.$$

Moreover $X_v(0) = Iv(0) \in \mathcal{H}$, because $F(0) = I$. Equations (19) and (20) give $\dashv(v, v) \ge \frac{1}{2}\|D_F v\|_2^2$. The actual inverse gives a polynomial conversion. If $I = \|F^{-1}\|_\infty$ and $C_1 = \|F_t\|_\infty$, then $v = F^{-1}D_F v - F^{-1}F_t F^{-1}X_v$, so $\|v\|_2 \le I(1+C_1 IT)\|D_F v\|_2$. Thus the Hilbert-space coercive inverse constructs the unique v with

$$\dashv(v,w) = -\int_0^T \langle f, X_w\rangle\, dt \quad \text{for every } w \in L^2(0,T;\mathcal{H}). \tag{22}$$

Set $B = Fv$.

Lemma 6.1 (Strong mean and its pressure)**.** *For a forcing f with continuous L^2 spatial derivatives of every order, the preceding construction gives continuous paths B and B_t in every spatial Sobolev space and a spatially smooth scalar q, whose spatial derivatives depend continuously on time locally in space, normalized by $q(t,0)=0$, such that*

$$B_t + MB + F^{-\top}\nabla_y q = f, \qquad F^{-1}B \in \mathcal{H}. \tag{23}$$

These identities hold pointwise, with one-sided time derivatives at both endpoints. In addition, for an actual $z_0 \in \mathcal{H}$,

$$B(0) = \mathcal{L}A_{\chi_\ell} z_0, \qquad \operatorname{supp} B(0) \subset \overline{B}(0,\ell^{-1}). \tag{24}$$

Proof. Translate the coefficients and forcing in the spatial variable. The coercive inverse is differentiable under each translation; differentiating its inverse identity expresses each new derivative using previously constructed derivatives and the known coefficient derivatives. This produces genuine L^2 derivatives, rather than a formal differentiation of a selected representative. The same argument applies to the translated cutoff operator, using its construction on the fixed homogeneous tensor space. Every finite derivative order is therefore available.

Let $Q_t : \mathcal{H} \to L^2$ be $Q_t z = F(t)z$. Its Gram operator $Q_t^* Q_t$ has a positive lower bound from F^{-1}. The weak equation, $F_{tt} = -HF$, and integration by parts give

$$v_t = (Q_t^* Q_t)^{-1} Q_t^* \{f - 2F_t v\}.$$

First this is an L^2 time identity. The H^1 trace construction and continuity of the coefficients then give continuous representatives of v and v_t, with the same spatial derivatives. The residual $r = f - 2F_t v - F v_t$ satisfies $Q_t^* r = 0$; equivalently, $F^\top r$ lies in the closed L^2 gradient space. Its smooth representative is curl-free; hence

$$q(t,y) = \int_0^1 \langle F(t,sy)^\top r(t,sy), y\rangle\, ds$$

satisfies $q(t,0) = 0$ and $\nabla q = F^\top r$. This proves (23). The pointwise equations follow from equality of continuous representatives of the same L^2 field. The Gram construction and closedness give $v(t) \in \mathcal{H}$ at every closed time. In particular, $B(0) = v(0) \in \mathcal{H}$. With $z_0 = Iv(0)$, the boundary identity in (22) is

$$P_\sigma\{-F_t(0)z_0 - B(0) + (M(0) + \mathcal{L}A_{\chi_\ell})z_0\} = 0.$$

Since $F_t(0) = M(0)$ and both $B(0)$ and $A_{\chi_\ell} z_0$ are solenoidal, this yields (24) as an L^2 identity. Spatial continuity gives the asserted pointwise support. $\square$

All these arguments are Hilbert-space arguments. For a transverse endpoint problem the map Q_t is replaced by $F(t)R_\perp$, where $R_\perp : \mathbb{R}^{d-1} \to m_0^\perp$ is a linear isometry onto the whole orthogonal complement. Its Gram lower bound is again supplied by F^{-1}. The energy, trace, and coercive-inverse arguments are unchanged; there are $d-1$ transverse variables throughout.

6.2 The joined high inverse and the full tensor corrector

Fix an activation time $0 < t_0 < T$. Choose an isometry $R_\perp : \mathbb{R}^{d-1} \to m_0^\perp$ onto the whole orthogonal complement and put $Q = FR_\perp$. Its range is $m(t,y)^\perp$, and $Q_t = MQ$, $Q_{tt} = -HQ$. The same fixed space $\mathbb{R}^{d-1}$ is used at every label and time; it does not discard any extra transverse direction.

For an angular-mean-zero forcing $f \in C([0,T]; H^q(\mathbb{R}^d \times \mathbb{T}_P; \mathbb{R}^d))$ for every finite q, first solve on $[0, t_0]$ the following displacement problem. Find $z \in H_0^1(0, t_0; L^2(\mathbb{R}^d \times \mathbb{T}_P; \mathbb{R}^{d-1}))$ such that

$$\int_0^{t_0} [\langle (Qz)_t, (Q\zeta)_t\rangle - \langle HQz, Q\zeta\rangle] dt = -\int_0^{t_0} \langle f, Q\zeta\rangle dt \quad (\zeta \in H_0^1). \tag{25}$$

Both displacement traces are zero. This condition does not prescribe the initial high velocity. Set $A = Qz_t$ on the history interval.

Lemma 6.2 (The bounded joined high inverse). *Under the known frame bounds and $Kt_0^2/2 \leq 1/2$, problem (25) has a unique solution. Its velocity A, its time derivative, and their spatial derivatives are continuous on $[0, t_0]$. At t_0 its actual velocity trace can be continued by the forward high equation. This gives a unique joined high operator on $[0,T]$ with the specified history boundary conditions. The history inverse and trace costs are polynomial in the known coefficient norms, inverse lower bound, t_0 and t_0^{-1}. No exponential forward estimate on $[0, t_0]$ is used.*

Proof. Write $X = Qz$. The one-sided Poincaré estimate and the curvature upper bound give

$$\int_0^{t_0} (|X_t|^2 - \langle HX, X\rangle) dt \geq \tfrac{1}{2}\|X_t\|_{L^2}^2.$$

If $Q^*Q \geq cI$, then $z = (Q^*Q)^{-1}Q^*X$, and direct differentiation, using $X_t = Q_t z + Qz_t$, gives the polynomial estimate

$$\|z_t\|_2 \leq c^{-1/2}(1 + t_0 c^{-1/2}\|Q_t\|_\infty)\|X_t\|_2.$$

Thus the form is coercive on the fixed Hilbert space of coordinate velocities z_t with zero time integral. Its bounded coercive inverse constructs z. Integration by parts and $Q_{tt} = -HQ$ give

$$(Q^*Q)z_{tt} = Q^*(f - 2Q_t z_t).$$

The actual positive Gram inverse therefore gives z_{tt} and the continuous H^1 traces. Repeated differentiation under spatial translation is justified on this same fixed Hilbert space, exactly as for the mean inverse. It gives all the stated derivatives and preserves both homogeneous displacement boundary conditions.

At order zero, a physical-space estimate avoids the frame conversion altogether. Put $M_* = \sup \|M\|$. Testing the form with z gives $\|X_t\|_2 \leq 2t_0\|f\|_2$. Since $A = X_t - MX$, it follows that

$$\|A\|_2 \leq 2t_0(1 + M_* t_0)\|f\|_2.$$

The velocity equation derived below implies $\|A_t\|_2 \leq 3M_*\|A\|_2 + \|f\|_2$. The elementary Hilbert-valued trace estimate consequently gives

$$\|A\|_{C_tL^2} \leq [2t_0(1 + M_* t_0)(t_0^{-1/2} + 3M_*\sqrt{t_0}) + \sqrt{t_0}]\|f\|_{L_t^2L^2}. \tag{26}$$

This also bounds the initial datum for the forward segment. □

The Gram equation and tangency show that the same A satisfies

$$A_t = \left(-M + 2\frac{m \otimes M^\top m}{|m|^2} \right) A + \left(I - \frac{m \otimes m}{|m|^2} \right) f. \tag{27}$$

Here $(a \otimes b)v = a\langle b, v\rangle$. Indeed the residual $f - A_t - MA$ is perpendicular to the whole range of Q, hence is a multiple of m. Differentiating $m \cdot A = 0$ gives its coefficient. For $t \geq t_0$ use the actual fundamental evolution of (27) and Duhamel's formula with datum $A(t_0)$. The velocities and their derivatives match at t_0 by the common right-hand side, so the joined field is strongly differentiable there as well as at the two closed endpoints.

Let g be a positive continuous function with $g(t_0) = 1$, and assume the supported forward bound

$$\|\mathcal{U}(t, s; y)\| \leq C_{\mathrm{f}}\, g(t)/g(s), \qquad t_0 \leq s \leq t \leq T, \quad y \text{ in the fixed support neighborhood.} \tag{28}$$

This hypothesis concerns the known parent evolution on the activation tail. The polynomial history estimate is supplied separately by (26).

Define the normalized periodic primitive ∂_θ^{-1} and put

$$\pi = \partial_\theta^{-1} \frac{\langle m, f\rangle - 2\langle m, MA\rangle}{|m|^2}. \tag{29}$$

Then $A_t + MA + m\partial_\theta \pi = f$. The inverse, trace, and forward maps commute with the angular integral. They therefore preserve zero angular mean. They also commute with multiplication by a fixed spatial cutoff, since these linear high equations contain no spatial derivative. Their uniqueness proves support preservation and covariance. Thus the joined high field and its pressure are smooth in (y, θ), periodic, and supported where the forcing is supported. Their spatial derivatives are continuous in time; the high velocity also has the continuous time derivative given by (27).

The primary uses a nonzero terminal displacement. To distinguish its physical and coordinate forms, let the selected physical endpoint be $Y_{\mathrm{phys}} \in m(t_0, 0)^\perp$, and fix once and for all

$$Y = R_\perp^* F(t_0, 0)^{-1} Y_{\mathrm{phys}} \in \mathbb{R}^{d-1}.$$

Then $Q(t_0, 0)Y = Y_{\mathrm{phys}}$ and $|Y| \leq \|F(t_0, 0)^{-1}\|\, |Y_{\mathrm{phys}}|$. This same coordinate vector is used at every label; the endpoint is not selected again as the label varies.

The actual cylinder problem has terminal datum $\alpha\chi_{\mathrm{in}}(y) f_\delta(\theta) Y$, which belongs to every cylinder Sobolev space, and initial displacement zero. Subtract the affine lift $\alpha(t/t_0)\chi_{\mathrm{in}} f_\delta Y$. Using $Q_{tt} = -HQ$, its remaining zero-trace problem is exactly (25) with forcing $-2\alpha t_0^{-1}\chi_{\mathrm{in}} f_\delta Q_t Y$. This forcing is continuous in every cylinder Sobolev space and has the prescribed compact spatial support.

Equivalently, at each fixed label solve the finite-dimensional homogeneous displacement problem with endpoints $0, Y$, and write $b_Y(t, y) = Qz_t$ for its velocity, continued homogeneously on the tail. The uniform frame bounds justify this pointwise solve and its label derivatives. Linearity and uniqueness identify the actual cylinder velocity with the compact periodic primary

$$A_1(t, y, \theta) = \alpha\chi_{\mathrm{in}}(y) f_\delta(\theta) b_Y(t, y). \tag{30}$$

Its initial value generally depends on y. The first stage, whose activation time is zero, instead uses a prescribed homogeneous forward datum. All subsequent algebra is identical. For higher grades the displacement endpoints in (25) are both zero, but their initial velocities are retained.

For every such A define all d^2 entries of the potential

$$Q_{ij} = \partial_\theta^{-1} \frac{m_i A_j - m_j A_i}{|m|^2}, \qquad Q_{ij} = -Q_{ji}, \qquad C_j = \sum_{i=1}^{d} D_y Q_{ij}[F^{-1} e_i]. \tag{31}$$

If $\kappa = k^{-1}$ and $\psi = \Phi_t^{-1}$, the chain rule gives

$$A_j(\psi(x), km_0 \cdot \psi(x)) + \kappa C_j(\psi(x), km_0 \cdot \psi(x)) = \kappa \sum_{i=1}^{d} \partial_{x_i} Q_{ij}(\psi(x), km_0 \cdot \psi(x)). \tag{32}$$

Indeed the phase part on the right is $\sum_i m_i \partial_\theta Q_{ij} = A_j$, by tangency. Taking divergence and interchanging the two ordinary derivatives cancels all terms by antisymmetry. This proof retains every pair (i, j), including mixed active and extra directions. Applying the actual volume-preserving Piola transformation shows that $F^{-1}(A+\kappa C)$ has zero divergence on every phase-shifted label graph. Integration over the shift gives its membership in the cylinder's closed divergence-free L^2 space. The potentials are smooth and have the same compact spatial support as A, so the integrations by parts are legitimate. The time derivative of C is obtained by differentiating (31): it includes the derivatives of F^{-1}, m, $|m|^{-2}$, the primitive, and A. This is the actual time derivative at the closed endpoints as well.

6.3 A single strict-prefix recursion

For vector fields a, b on the time-label-angle cylinder write

$$S(a, b) = D_y b[F^{-1} a], \qquad T(a, b) = \langle m, a \rangle \partial_\theta b, \qquad La = \partial_t a + Ma.$$

Let A_1 be the actual homogeneous joined primary (30) (or the first-stage forward primary). Put $B_1 = q_1 = 0$, form C_1 by (31), and set every grade-zero field to zero. Suppose grades below $p \geq 2$ have been constructed. Set

$$V_i = A_i + B_i + C_{i-1} \quad (1 \leq i < p), \qquad \widehat{V}_p = C_{p-1},$$

and extend this known sequence by zero above p. The literal known force is

$$K_p = -LC_{p-1} - F^{-\top} \nabla_y \pi_{p-1} - \sum_{i+j=p} S(\widehat{V}_i, \widehat{V}_j) - \sum_{i+j=p+1} T(\widehat{V}_i, \widehat{V}_j). \tag{33}$$

Every term uses only the strict prefix. Let an overline denote normalized angular averaging. First solve the actual mean problem with forcing $\overline{K}_p$, obtaining B_p, q_p. Then apply the joined high inverse of Lemma 6.2 with forcing

$$f_p = K_p - \overline{K}_p - \langle m, B_p \rangle \partial_\theta A_1, \tag{34}$$

obtaining A_p, π_p, and form C_p from A_p. The final term of (34) has zero angular mean. Terms in K_p containing only means are angularly constant and disappear from $K_p - \overline{K}_p$. Every remaining term contains a high field or corrector, so it has the same compact spatial support as the seed. The actual mean forcing has all continuous L^2 spatial derivatives by the Sobolev product estimates. The high forcing is smooth, periodic, mean zero, and spatially supported. These facts inductively verify the domains of the two solvers.

There is no simultaneous nonlinear solve at grade p. In the quadratic coefficient, replacing $\widehat{V}_p$ by $\widehat{V}_p + A_p + B_p$ adds only $\langle m, B_p \rangle \partial_\theta A_1$. The other potentially new fast terms vanish because A_p, A_1 are tangent and B_p is angularly constant. This proves both strict-prefix coherence and the grade-p momentum equation. In particular, families constructed for different finite cutoffs are restrictions of one family.

6.4 Finite truncation, divergence, and the residual identity

For $N \geq 1$ define

$$V^{[N]} = \sum_{i=1}^{N} \kappa^i (A_i + B_i + \kappa C_i),$$
$$p^{[N]} = \sum_{i=1}^{N} \kappa^i (q_i + \kappa \pi_i). \tag{35}$$

Equivalently the velocity grades are $V_i = A_i + B_i + C_{i-1}$ for $1 \leq i \leq N$, followed by the terminal grade $V_{N+1} = C_N$. The pressure grades are $p_i = q_i + \pi_{i-1}$ for $1 \leq i \leq N$, followed by $p_{N+1} = \pi_N$. Grades outside this range are zero. In particular the terminal grade does not include unsolved A_{N+1} or B_{N+1}.

Each high-corrector pair in (35) is exactly solenoidal after graph evaluation by (32). Each mean has $F^{-1}B_i \in \mathcal{H}$ by Lemma 6.1; its constant-angle embedding therefore belongs to the same cylinder solenoidal space. Thus the complete finite sum satisfies the closed divergence constraint. The constant-angle embedding of an ordinary L^2 gradient lies in the cylinder's closed gradient space: approximate it by gradients of compactly supported smooth scalars and use $\nabla_\kappa \varphi = \kappa \nabla_y \varphi$ for an angle-independent scalar. This applies to the mean pressure gradients. The compact periodic high scalar pressures themselves give gradient generators. Their finite sum therefore belongs to the required closed gradient space.

For any raw V, p define its label momentum residual by

$$\mathcal{R}_\kappa(V,p) = LV + F^{-\top}\nabla_y p + \kappa^{-1} m \partial_\theta p + S(V,V) + \kappa^{-1} T(V,V). \tag{36}$$

For the finite grade sequences above, let

$$R_n = LV_n + F^{-\top}\nabla_y p_n + m\partial_\theta p_{n+1} + \sum_{i+j=n} S(V_i,V_j) + \sum_{i+j=n+1} T(V_i,V_j).$$

The actual mean and high equations give $R_n = 0$ for $0 \leq n \leq N$. Expanding the finite sums with their actual derivatives, including the one-sided time derivatives, consequently gives the exact identity

$$\mathcal{R}_\kappa(V^{[N]}, p^{[N]}) = \sum_{n=N+1}^{2N+2} \kappa^n R_n. \tag{37}$$

Neither a time derivative of p nor an infinite formal expansion is used. In particular, R_{N+1} retains LC_N and $F^{-\top}\nabla_y \pi_N$.

6.5 Quantitative bounds at one known parent

We give the estimates in a form that makes the derivative counting explicit. For an integer $a \geq 0$ set $\mathcal{G}_a(j) = R^{j+a}((j+a)!)^2$. Use ordered cylinder derivatives with alphabet $\{1, \ldots, d+1\}$. More precisely, the block of order j is

$$\mathrm{Block}_j(u) = \sum_{|w|=j} \sum_{l=0}^{b_d} \sum_{|v|=l} \|\partial^v \partial^w u\|_{C([0,T];L^2)}.$$

At fixed time the corresponding word Sobolev norm is equivalent to the usual H^{b_d} norm, with fixed constants depending only on d, b_d. A positive time weight is inserted by dividing the actual field by that weight before taking spatial or angular derivatives. For one sufficiently large R, depending only on the known coefficients, seed, and solver bounds, the preceding recursion gives

$$\begin{aligned} \mathrm{Block}_j(B_p, B_{p,t}, \nabla_y q_p) &\le H_0^{2p-2}\mathcal{G}_{100p-140}(j), && p \ge 2, \\ \mathrm{Block}_j(A_p, A_{p,t}, C_p, C_{p,t}, \nabla_y \pi_p) &\le \gamma(t) H_0^{2p-2}\mathcal{G}_{100p-80}(j), && p \ge 1. \end{aligned} \tag{38}$$

Here each entry has the stated bound separately. We use $\gamma(t) = \alpha$ for $0 \le t \le t_0$ and $\gamma(t) = \alpha g(t)$ on the tail, with $H_0 \ge \max(1, \sup \gamma)$. The notation with $\gamma(t)$ means the uniform-time block of the field divided by γ. The supported forward constant in (28), the history trace cost, and the actual terminal datum are included among the known source constants controlling R. In particular the higher initial high fields are bounded with the factor α, not set to zero. This is the family used below and in Section 9.

For completeness, the derivative estimates underlying (38) are as follows. The word Leibniz rule is bounded by a binomial convolution. Its Gevrey convolution closes because

$$\binom{j}{l}(l!)^2((j-l)!)^2 = \frac{(j!)^2}{\binom{j}{l}}, \qquad \sum_{l=0}^{j}\binom{j}{l}^{-1} \le 3.$$

Fixed-order Sobolev products add a constant depending on b_d, d. On the history interval, differentiate the actual fixed-space coercive inverse and then use the Gram equation and its trace. On the tail, differentiate the actual Duhamel equation and use (28). The history terminal trace is the datum in this Volterra recurrence. The joined high velocity and its time derivative consume at most three grades. Differentiating the actual coercive inverse, followed by its Gram reconstruction and the H^1 time trace, consumes at most three grades for the mean, its time derivative, and its pressure gradient. These are estimates on word blocks with the full alphabet. The angular mean is a bounded L^2 map with norm $P^{-1/2}$, and the constant-angle embedding has norm $P^{1/2}$; the factors cancel. Thus converting mean input to mean output does not multiply the unknown word radius at each grade. Only the known coefficient radius is enlarged once. Also the mean time weight is the constant H_0^{2p-2}, so no commutation of a time-dependent weight with the mean inverse is claimed. The history estimate is applied to f/α and the tail estimate to $f/(\alpha g)$; linearity preserves the common factor α. The tail uses the weighted Duhamel operator and never differentiates g. The tensor potential and its actual time derivative cost one spatial derivative and a fixed d-dependent multiplier constant; the pressure source (29) and one slow derivative cost at most two more grades. Enlarging R once absorbs these fixed constants. The grade counting can be made explicit. The mean force has shift at most $100p - 150$, and the high force has shift at most $100p - 90$. For example a slow high–high term with $i + j = p$ has shift $100p - 159$. In a fast term with $i + j = p + 1$, the first high factor is annihilated by $m \cdot A_i = 0$. A first corrector factor has shift $100i - 180$, so the resulting high term has shift at most $100p - 159$; a first mean factor gives shift $100p - 119$, and has zero angular average. The new term $(m \cdot B_p)\partial_\theta A_1$ has the latter shift as well. These leave the displayed force margins. The profile powers close because $\gamma \le H_0$; in particular a mean–high fast product has exactly the allowed factor γH_0^{2p-2}. The at most quadratic number of summands is absorbed by the spare factorial shifts: for $a \ge 0$, $\mathcal{G}_{a+2}(j)/\mathcal{G}_a(j) = R^2(j + a + 1)^2(j + a + 2)^2$. The low grades are covered by one enlargement of the known radius. The mean inverse and the joined high inverse, followed by the corrector and pressure estimates, use fewer than the ten reserved shifts. Strong induction proves (38) for this same joined family, without an estimate on a yet unconstructed output.

For the correction use the truncated weighted norm

$$\|u\|_{r,b_d;J} = \sum_{j=0}^{J}\frac{r^j}{(j!)^2}\sum_{|w|=j}\sum_{l=0}^{b_d}\sum_{|v|=l}\|\partial^v\partial^w u\|_2.$$

The final conversion from a block bound $C\mathcal{G}_a(j)$ costs at most $2C(4R)^a(a!)^2$ when $4Rr \le 1/2$. This conversion is performed after the word recursion, not at every recursive step. In particular, at every derivative cutoff,

$$\|V_i\|_{r,b_d;J} \le 6H_0^{2i}(4R)^{100i}((100i)!)^2 \quad (i \ge 1),$$

with analogous bounds for the actual time derivatives and pressure forces at a smaller fixed radius. For explicit expansion-order counting, choose a known-source constant $C_0 \ge 1$ and put $S_N = C_0(N+1)^{200}$. The inequality $(100i)! \le (100(N+1))^{100i}$ for $i \le N+1$ bounds each displayed grade by S_N^i after enlarging C_0. The actual time derivatives and pressure terms obey the same bound; one slow or angular derivative is paid by one fixed radius reduction, whose reciprocal cost is included in C_0. A slow product with $i+j=n$ costs at most CS_N^n, and a fast product with $i+j=n+1$ at most CS_N^{n+1}. There are at most $(N+2)^2$ terms. The terminal linear terms have the same bound, including the true C_N time derivative and the slow derivative of π_N. Thus $C_1(N+2)^2S_N^{n+1}$ bounds the entire residual grade. Taking $B_N = C_2(N+1)^{202}$ with a sufficiently large fixed C_2 absorbs this factor for every $n \ge N+1$. The convenient larger exponent 300 gives

$$B_N \le C_*(N+1)^{300}, \qquad \|R_n\|_{r,b_d;J} \le B_N^{n+1} \quad (N+1 \le n \le 2N+2). \tag{39}$$

The constant C_* is fixed by the current parent and seed, and is independent of N, k, J. Spatial derivatives of the high pressure, the time derivative of C_N, and the terminal linear terms are all included in this estimate.

Put

$$Z = kF^{-1}V^{[N]}, \quad Q^a = k^2p^{[N]}, \quad \nabla_\kappa = \kappa\nabla_y + m_0\partial_\theta, \quad G = F^{-1}F^{-\top}. \tag{40}$$

Then the actual coordinate residual is $\mathcal{E} = kF^{-1}\mathcal{R}_\kappa(V^{[N]}, p^{[N]})$. If $B_N \le k^{1/100}$ and the known inverse multiplier cost is at most $k^{1/100}$, the geometric sum in (37) yields, for $N \ge X-1$, $X \ge 6$,

$$\|\mathcal{E}\|_{r,b_d;J} \le \exp(-\tfrac{7}{10}X\log k). \tag{41}$$

The field Z and its first cylinder derivatives have fixed weighted bounds B_0, B_1. More sharply, its actual cylinder transport vector is

$$\mathcal{V}_\kappa Z = (\kappa Z, m_0 \cdot Z), \qquad \|\mathcal{V}_\kappa Z\|_{r,b_d;J} \le D_*/k.$$

The order-one high field is tangent, so its fast transport component vanishes exactly. All other contributions carry at least one inverse frequency. This fact is essential: using the full $O(1)$ norm of Z in place of its small transport norm would lose the needed radius bound.

6.6 An exact correction on the cylinder

The coordinate equation is

$$W_t + (W\cdot\nabla_\kappa)W + 2F^{-1}F_tW + \kappa\sum_{i=1}^{d} W_iF^{-1}(\partial_iF)W + G\mathbf{p} = 0, \qquad \nabla_\kappa\cdot W = 0. \tag{42}$$

Here $\mathbf{p}$ is a vector pressure variable in $\mathcal{G}_\kappa = \overline{\{\nabla_\kappa\varphi : \varphi \text{ smooth, periodic, compact in } y\}}^{L^2}$. The notation does not assert that $\mathbf{p}$ is the derivative of a global periodic scalar. In particular no closed-range assertion for the degenerate d-component operator on the $(d+1)$-dimensional cylinder is used. All d quadratic coefficient matrices occur in this equation. Substituting Z and the actual approximate pressure vector $\mathbf{p}^a = \nabla_\kappa Q^a$ gives exactly the residual $\mathcal{E}$ above.

The correction pressure is solved in this closed subspace of vector-valued cylinder L^2. The positive matrix G gives a coercive variational pressure inverse on that subspace. Its derivatives are controlled by the true coefficient jets and commutators.

Seek $W = Z + E$ with $E(0) = 0$. Use the inverse metric $G^{-1} = F^\top F$ in the finite weighted Sobolev energy. The top pressure term cancels since $\langle G^{-1}E, G\mathbf{p}^e\rangle = \langle E, \mathbf{p}^e\rangle = 0$ for $\mathbf{p}^e \in \mathcal{G}_\kappa$. The top transport term is integrated by parts using the closed solenoidal constraint. Differentiated metric terms and lower products have a known coefficient bound. To specify the energy convention, put $\|u\|_{F^\top F} = (\int\langle F^\top F u, u\rangle)^{1/2}$ and set

$$\mathcal{A}_J = \sum_{j=0}^{J} \frac{r^j}{(j!)^2} \sum_{|w|=j} \sum_{l=0}^{b_d} \sum_{|v|=l} \|\partial^v \partial^w E\|_{F^\top F}.$$

Thus $\mathcal{A}_J$ is a sum of metric norms, not a squared norm; it is uniformly equivalent to the preceding weighted norm by the known frame bounds. Let $\mathcal{L}_J$ be the same sum with the additional factor j in every summand. Derivatives of norms at zero can be justified by first replacing each norm by its quadratic regularization and then letting the regularization parameter decrease to zero.

The orthogonal gradient projection commutes with every cylinder derivative. In the differentiated pressure equation, each nontrivial commutator with G lowers the derivative order on the pressure by at least one. This compensates for the one derivative in the transport source. After the fixed-base Sobolev pressure estimate, its remaining coefficient-word convolutions are absorbed by choosing the known radius small enough. Thus no unknown derivative of order $J+b_d+1$ is left in the energy bound. The word product convolution and the metric transport cancellation consequently give a single radius-loss factor. In fact, in a transport commutator with $1 \le j \le n$, put $l = n-j+1$. The ratio of Gevrey weights, including its Leibniz coefficient, is

$$\frac{r^n}{(n!)^2}\binom{n}{j}\frac{(j!)^2(l!)^2}{r^j r^l} = \frac{l^2}{r\binom{n}{j}} \le \frac{l}{r},$$

since $\binom{n}{j} \ge l$. The excluded term $j = 0$ is precisely the cancelled principal transport term. Thus the loss is the factor l in $\mathcal{L}_J$, rather than an uncontrolled factor l^2 or an extra derivative beyond the cutoff. We obtain

$$\mathcal{A}_J' \le C\{\mathcal{A}_J + \mathcal{A}_J^2 + \varepsilon\} + \left\{\frac{r'}{r} + C(r^{-1} + R_c)(D_*/k + \mathcal{A}_J)\right\}\mathcal{L}_J. \tag{43}$$

The same C, R_c work for every external derivative cutoff J. Here $\varepsilon = 2\exp(-7X\log k/10)$ includes the metric conversion of the residual after a fixed enlargement of constants.

Choose a positive r_0 from the known coefficient and packet radii so that all pressure absorptions hold, in particular $r_0 R_c \le 1$, and set

$$\eta = \exp(-\sqrt{X}), \qquad r(t) = r_0 - 2C(D_*/k + \eta)t.$$

If

$$2\varepsilon e^{3CT} \le \eta/2, \qquad 2C(D_*/k+\eta)T \le r_0/2, \tag{44}$$

then $r_0/2 \le r(t) \le r_0$. In the bootstrap range $\mathcal{A}_J \le \eta \le 1$ the loss coefficient in (43) is nonpositive. The scalar integral inequality gives the strict bound $\mathcal{A}_J \le \eta/2$ and closes the bootstrap.

We give the existence step explicitly, following the pressure projection and vanishing-viscosity argument of [1, §3.6]. Let $\mathcal{G}_\kappa$ be the closed gradient space above and let $P_\mathcal{G}$ be its orthogonal projection in cylinder L^2. On $\mathcal{G}_\kappa$ the map $g \mapsto P_\mathcal{G}(G(t)g)$ is coercive, since $\langle g, G(t)g\rangle \ge c\|g\|_2^2$ with $c > 0$ supplied by F. Set

$$\mathcal{T}_t = (P_\mathcal{G} G(t)|_{\mathcal{G}_\kappa})^{-1} P_\mathcal{G}, \qquad \mathcal{P}_t = I - G(t)\mathcal{T}_t.$$

Then $P_{\mathcal{G}}\mathcal{P}_t = 0$, $\mathcal{P}_t u = u$ for $u \in \mathcal{G}_\kappa^\perp$, and $\mathcal{P}_t G(t)g = 0$ for $g \in \mathcal{G}_\kappa$. Differentiating the coercive pressure equation gives the Sobolev bounds for $\mathcal{T}_t$ and $\mathcal{P}_t$ at every fixed order. Their constants may depend on that fixed order; the previously derived weighted energy constants do not depend on the external derivative cutoff.

Write the unprojected correction source as

$$\mathcal{N}(E) = b_E \cdot \nabla_{y,\theta} E + \mathcal{Z}(E) + \mathcal{E}, \qquad b_E = \mathcal{V}_\kappa(Z + E),$$

where $\mathcal{Z}(E)$ contains the terms $E \cdot \nabla_\kappa Z$, $2F^{-1}F_t E$ and the differences of the two quadratic coefficient terms. It has no derivative of E. For $0 < \nu \leq 1$, solve

$$E_t^\nu - \nu \Delta_{y,\theta} E^\nu = -\mathcal{P}_t \mathcal{N}(E^\nu), \qquad E^\nu(0) = 0. \tag{45}$$

The right side is locally Lipschitz $H^q \to H^{q-1}$ for $q \geq b_d$, and the Fourier heat estimate

$$\|e^{\nu t \Delta} f\|_{H^q} \leq C_q (1 + (\nu t)^{-1/2}) \|f\|_{H^{q-1}}$$

gives a local mild solution by contraction. Every component of ∇_κ has constant coefficients and commutes with the full cylinder Laplacian. The heat semigroup therefore preserves the fixed space $\mathcal{G}_\kappa^\perp$. Since the non-heat term in (45) takes values in this space, the mild solution is exactly divergence free. No time derivative of $\mathcal{P}_t$ and no commutation of $\mathcal{P}_t$ with the heat semigroup is required.

The actual pressure gradient is $-\mathcal{T}_t \mathcal{N}(E^\nu)$. The viscosity contributes no pressure, because ΔE^ν is still divergence free. Heat maximal regularity justifies testing differentiated equations by $F^\top F \partial^w E^\nu$: on each compact existence interval, $E^\nu \in L_t^2 H^{q+1}$ and $E_t^\nu \in L_t^2 H^{q-1}$. The differentiated metric-viscosity term satisfies

$$\nu \int \langle F^\top F u, \Delta u \rangle \leq -\tfrac{c\nu}{2} \|\nabla u\|_2^2 + C\nu \|u\|_2^2.$$

This follows by integration by parts and Young's inequality, using the known first derivatives of $F^\top F$. Thus it adds only a bounded multiple of the lower energy to (43), uniformly for $\nu \leq 1$. In the choice of C above we include this known metric-viscosity constant before selecting the radius or frequency; it depends only on the fixed source and not on J or ν. The same norm regularization justifies summing this estimate for $\mathcal{A}_J$. The bootstrap extends every viscous solution to T. Uniqueness in the low Sobolev class identifies the solutions constructed at different higher orders. In particular they have uniform bounds in $C_t H^q$ for each finite q.

There is no compactness assumption at spatial infinity in removing viscosity. For two viscosities, subtract their equations and use the same metric L^2 difference energy. The transport and pressure leading terms cancel as before. The remaining viscous defect is $(\nu - \mu)\Delta E^\mu$, uniformly bounded in L^2 by a constant of the fixed source times $|\nu - \mu|$. With zero initial difference, Gronwall gives

$$\|E^\nu - E^\mu\|_{C_t L^2} \leq C_* |\nu - \mu|.$$

The dissipative term $\nu \Delta(E^\nu - E^\mu)$ is handled by the same integration by parts. Interpolation with the uniform higher-order bounds now gives a strong limit in $C_t H^q$ for every finite q. The nonlinear source and the actual pressure inverse pass to this limit one order lower; $\nu \Delta E^\nu \to 0$ there. The gradient and solenoidal constraints are closed, the finite weighted energies pass to the limit, and the initial trace remains zero. The limiting equation supplies the continuous strong time derivative at every Sobolev order, including the within-interval endpoint derivatives.

These steps use the heat Fourier multiplier on $\mathbb{R}^d \times \mathbb{T}_P$, a closed Hilbert gradient space, and the Sobolev product threshold (18). In the three-dimensional proof the corresponding cylinder has dimension four; here the only changes are the $d + 1$ Fourier variables, all d

quadratic coefficient matrices, and the dimension-dependent Sobolev constants. Neither a curl identity nor a two-dimensional transverse reduction enters this existence argument.

There is an explicit finite frequency choice. With

$$X = k^{1/301000}, \qquad N = \lfloor X \rfloor,$$

(39) meets $B_N \le k^{1/100}$ at sufficiently large k. More concretely, after denoting by I_* the inverse-multiplier cost, it is enough for the packet and the correction guards that

$$k \ge \left(64 + e + 2^{300}C_* + I_* + 12CT + \frac{8CTD_*}{r_0} + \left(\frac{8CT}{r_0}\right)^2\right)^{301000}. \tag{46}$$

Indeed the 301000th root controls X, the polynomial tail, $\log k$, and the two terms in the radius loss; the negative exponential in (41) dominates e^{3CT} and $e^{-\sqrt{X}}$. The constants in (46) are quantities of the current known source. This statement gives no bound for them in terms of the parameters of a future parent.

6.7 The physical child and the signed pressure term

Define the child by

$$U^+(t,x) = U(t,x) + \kappa F(t,y) W(t,y,km_0 \cdot y), \qquad y = \Phi_t^{-1}(x). \tag{47}$$

The chain rule, $\Phi_t' = U \circ \Phi_t$, and (16) transform (42) into the physical momentum equation with pressure force

$$\nabla P_U(t,x) + \kappa F(t,y)^{-\top} \mathbf{p}(t,y,km_0 \cdot y).$$

This force is an actual gradient on physical space. To see this without assuming a cylinder scalar potential, observe that closed-gradient membership implies

$$D_{\kappa,i}\mathbf{p}_j - D_{\kappa,j}\mathbf{p}_i = 0$$

in distributions: it holds for each compact smooth gradient generator and is preserved by the L^2 limit. All spatial derivatives of the constructed pressure vector are continuous, so the same identity holds pointwise. The pullback chain rule for a one-form gives

$$\partial_{x_i}(\kappa F^{-\top}\mathbf{p})_j - \partial_{x_j}(\kappa F^{-\top}\mathbf{p})_i = \sum_{a,b} (F^{-1})_{bi}(F^{-1})_{aj}(D_{\kappa,b}\mathbf{p}_a - D_{\kappa,a}\mathbf{p}_b) = 0$$

after graph evaluation. The terms differentiating $F^{-1} = D\Phi_t^{-1}$ cancel by the symmetry of its ordinary second derivatives.

Graph evaluation uses the smooth periodic representatives and one extra angular Sobolev trace estimate; the parent change of variables is volume preserving. It gives all physical L^2 spatial derivatives and their continuous time paths, both for the velocity and for this pressure force. A curl-free L^2 vector field on $\mathbb{R}^d$ lies in the ordinary closed gradient space, by its Fourier characterization. The actual Piola identity also transforms the closed cylinder solenoidal constraint into $\operatorname{div} U^+ = 0$. Taking the ordinary divergence of the physical momentum equation and applying the Helmholtz decomposition identifies its pressure force with the gradient of the Newton pressure in Lemma 2.2. We choose exactly that normalized scalar pressure. The smooth Sobolev local theory then identifies the constructed field with the actual Euler evolution on the whole interval. Auxiliary mean and finite angular scalars fixed convenient representatives, but no global periodic scalar for the correction was required.

The same field has the decomposition

$$U^+ = U + V^{[N]}(t,y,km_0 \cdot y) + \kappa F(t,y) E(t,y,km_0 \cdot y).$$

Thus its initial increment is exactly the finite packet, and the correction initially vanishes. The all-order correction estimate gives physical velocity-gradient and pressure-Hessian errors bounded by fixed current-source constants times η. One fast derivative of the velocity correction is paired with its prefactor κ. The physical pressure force of the correction is $\kappa F^{-\top}\mathbf{p}^e$ evaluated on the graph. Its one physical derivative, which is the pressure Hessian, has the same cancellation of κ against one fast derivative. For the finite scalar primary pressure the equivalent accounting is its prefactor κ^2 against two fast derivatives. All remaining powers of κ are nonnegative. A fixed number of the uniform weighted estimates controls the required coefficient and angular derivatives; the stated constants are independent of k.

The periodic profile in (30) can be chosen with explicit polynomial dependence on its width. Fix an even Gevrey–2 function η supported in $[-1,1]$, with $0 \le \eta \le 1$ and $\eta(0) = 1$, and write $m_\eta = \int_{\mathbb{R}} \eta > 0$. For example, take $\eta(s) = \exp(1-(1-s^2)^{-1})$ on $|s| < 1$ and zero elsewhere. On a complex disc of radius $c(1-s^2)$ about a real $s \in (-1,1)$, its modulus is at most $C\exp(-c'/(1-s^2))$. Cauchy's estimate, followed by maximizing $t^{-n}e^{-c'/t}$, yields $\|\eta^{(n)}\|_\infty \le CR^n(n!)^2$; the same estimate gives the zero derivative traces at ± 1. Choose a fixed $a > 0$ with $a \le P/4$ and $am_\eta \le P/4$. For $0 < \delta \le 1$, let $w = a\delta$, let η_w denote the P-periodization of $\eta(\theta/w)$, and put $q_w = wm_\eta/P$. Define f_δ by

$$f_\delta(0) = 0, \qquad f_\delta'(\theta) = \frac{\eta_w(\theta) - q_w}{\delta(1-q_w)}.$$

The derivative is even and has zero angular mean, so this defines an odd P-periodic function with zero mean. Since $0 < q_w \le 1/4$,

$$f_\delta'(0) = \delta^{-1}, \qquad -\frac{1}{3} \le f_\delta' \le \delta^{-1}.$$

Indeed $q_w/\delta = am_\eta/P \le 1/4$. If $\|\eta^{(n)}\|_\infty \le C_0R_0^n(n!)^2$, differentiating this formula gives, for constants C_*, R_* depending only on the fixed bump and period,

$$\|f_\delta^{(n)}\|_\infty \le C_*(R_*/\delta)^n(n!)^2 \quad (n \ge 0).$$

At order zero, integrate $|f_\delta'|$ over a period: its positive and negative parts have equal integrals and each is bounded by a constant independent of δ. For $n \ge 1$, the factors are $\delta^{-1}w^{-(n-1)}$ and the derivatives of the fixed bump. Thus the profile's Gevrey radius is a polynomial source cost.

Use this profile for the joined primary (30). The uncut physical high velocity is b_Y, and the amplitude is α. At $y = \Phi_t^{-1}(x)$ and $\vartheta = km_0 \cdot y$, the leading velocity and pressure Hessian tensors are respectively

$$\begin{aligned} \mathcal{S} &= \alpha\chi_{\mathrm{in}}(y)f_\delta'(\vartheta)\, b_Y \otimes m, \\ \mathcal{Q} &= -2\alpha\chi_{\mathrm{in}}(y)\frac{\langle m, Mb_Y\rangle}{|m|^2}f_\delta'(\vartheta)\, m \otimes m. \end{aligned} \tag{48}$$

The actual finite higher grades, derivatives of slow coefficients, and exact correction give

$$\|DU^+ - DU - \mathcal{S}\| \le C_v/k + C_v'\eta, \quad \|D^2P_{U^+} - D^2P_U - \mathcal{Q}\| \le C_p/k + C_p'\eta, \tag{49}$$

uniformly on the closed time interval and physical space. The pressure identity follows first for the true finite scalar pressure and then for the true correction pressure; equality of their gradient with the canonical Euler pressure gradient transfers the identity to P_{U^+}.

If $\langle m, Mb_Y\rangle \ge 0$, the lower slope bound gives the one-sided estimate

$$\langle \mathcal{Q}z, z\rangle \le 2\alpha\chi_{\mathrm{in}}\langle m, Mb_Y\rangle|z|^2. \tag{50}$$

Without that sign the valid absolute estimate is

$$\|\mathcal{Q}\| \leq 2\alpha \|M\| \chi_{\text{in}} |m| \, |b_Y| / \delta.$$

The distinction is necessary: using the absolute bound at late times would lose the narrow-profile gain in the next mean curvature budget. All tensors in (48) act on the entire $\mathbb{R}^d$, and the norm identities used here hold in every dimension.

For spatial localization the preceding construction is applied to the actual rescaled parent

$$U^\rho(t,x) = \rho^{-1} U(t, \rho x), \qquad P_{U^\rho}(t,x) = \rho^{-2} P_U(t, \rho x), \qquad 0 < \rho \leq 1.$$

Direct substitution proves the Euler equations on the same time interval. Its actual flow is $\rho^{-1}\Phi_t(\rho y)$, so its frame, strain and pressure Hessian in labels are exactly $F(t, \rho y)$, $M(t, \rho y)$ and $H(t, \rho y)$. Undoing the scaling after constructing the child gives the literal physical field $\rho(U^\rho)^+(t, x/\rho)$. Velocity gradients and pressure Hessians are unchanged in size by these two inverse scalings, and the support radius of a compact oscillatory coefficient is multiplied by ρ. Thus the low error bounds and leading tensors above transfer to the original parent with no factor ρ^{-1} in those two derivative orders. Higher-order initial derivatives retain their true scaling factors as recorded in Section 9.

6.8 Physical localization of the mean boundary condition

The oscillatory localization scale and the mean cutoff scale are chosen separately. With a fixed physical cutoff parameter ℓ_* and physical unscaling $x = \rho y$, use the normalized cutoff parameter $\ell = \rho \ell_*$. Then $\chi(\ell y) = \chi(\ell_* x)$, and (24) places each physical mean initial increment in $\overline{B}(0, \ell_*^{-1})$. If $\operatorname{supp} \chi_{\text{in}} \subset \overline{B}(0, 1/4)$, the high and corrector initial increments are supported in $\overline{B}(0, \rho/4)$. The exact correction has initial value zero. Lemma 7.1 proves propagation of the mean reserve with this fixed physical cutoff, and Lemma 7.2 gives the resulting common compact support through the iteration.

6.9 The one-parent packet conclusion

We collect the output needed by the iteration. The assumptions in the following lemma concern only the current, already known source.

Lemma 6.3 (Exact joined packet child). *Fix $d \geq 3$ and a smooth Sobolev parent on $[0, T]$ with actual flow* (16)*, the known coefficient bounds* (17)*, and the mean reserve* (20)*. Fix its transverse terminal datum, compact Gevrey cutoffs, and the positive forward profile with* (28)*. There is a finite constant determined by these data such that every sufficiently large k gives the actual child* (47) *from the joined strict-prefix family. It solves the full d-dimensional Euler equation with its Newton pressure on the whole closed interval and belongs to $C_t H^m \cap C_t^1 H^{m-1}$ for every integer $m \geq 1$.*

Its leading velocity-gradient and pressure-Hessian increments are exactly the two tensors in (48)*. After an explicit enlargement of the known-source frequency threshold, each error in* (49) *is at most $k^{-1/4}$, uniformly in time and space. Its initial increment is the entire finite sum* (35) *at time zero, including every higher high initial velocity and the terminal corrector C_N; the exact correction has initial value zero.*

If the parent is odd and the cutoffs are even, then the child is odd. If in addition a fixed orthogonal coordinate reflection $\mathcal{R}$ obeys

$$U(t, \mathcal{R}x) = \mathcal{R}U(t,x), \quad \mathcal{R}m_0 = m_0, \quad \mathcal{R}R_\perp Y = R_\perp Y,$$

and preserves the cutoffs, then

$$U^+(t, \mathcal{R}x) = \mathcal{R}U^+(t,x), \qquad DU^+(t, \mathcal{R}x) = \mathcal{R}DU^+(t,x)\mathcal{R}^{-1}.$$

For the fixed physical mean cutoff described above, every initial increment is supported in $\overline{B}(0, \max(1/4, \ell_*^{-1}))$. *Thus the child's initial support is contained in the union of this ball and the parent's initial support.*

Proof. The constructions and identities above give existence, the actual PDE, the normalized pressure, and all Sobolev orders. The terminal corrector and pressure were included before computing the literal residual, so no additional endpoint grade is omitted.

Here is a concrete enlargement giving the stated error. Let $\beta = 301000$ and let $C_{\mathrm{err}} \geq 1$ dominate the four constants in (49). Increase the lower bound for $X = k^{1/\beta}$ to include $(2\beta)^4$ and $(4C_{\mathrm{err}})^2$, as well as the quantity inside parentheses in (46). For $X \geq 1$, $\log X \leq 4X^{1/4}$, and hence

$$\tfrac{1}{4}\log k \leq \tfrac{1}{2}\sqrt{X}, \qquad \log(4C_{\mathrm{err}}) \leq \tfrac{1}{2}\sqrt{X}.$$

It follows that $C_{\mathrm{err}}e^{-\sqrt{X}} \leq k^{-1/4}/4$ and $C_{\mathrm{err}}/k \leq k^{-1/4}/4$. The sum satisfies the asserted error bound. The enlarged threshold remains a fixed power of a sum of known source constants.

For symmetry, uniqueness of the parent's flow first gives $\Phi_t(-y) = -\Phi_t(y)$ and the corresponding reflection covariance. The mean form commutes with the induced orthogonal action, since its cutoff tensor operator is constructed using all tensor indices and the cutoff is invariant. Uniqueness of its coercive inverse gives the mean parities. For the joined high operator the action preserves the fixed zero-trace displacement space; for the primary it also preserves the terminal datum. Uniqueness of the history inverse and then of the forward initial-value equation proves the high parities. The normalized angular primitive, the full tensor corrector, and each literal product in the strict-prefix force commute with the same actions. Induction therefore gives odd vector fields and even scalar pressures for the whole finite family, together with the stated reflection covariance.

The projected coordinate equation and its closed solenoidal and gradient spaces are also invariant under these actions. Its unique correction with zero initial value has the same parities. Transforming back gives the actual child symmetry; differentiating this identity gives the claimed full derivative covariance. Finally, the initial support follows from (24), support preservation of the joined high inverse and the tensor potential, and $E(0) = 0$. $\square$

7 A fixed compact support and the full mean reserve

The initial mean boundary cutoff and the oscillatory packet have different spatial scales. In physical coordinates fix two parameters $0 < \ell_* \leq 1$ and $0 \leq r_* \leq 1/4$. Write c_d for the constant in the localized mean boundary inequality. Let L_* bound all time horizons. The correct quantity to propagate is

$$\mathcal{M} = \tfrac{1}{2}KL_*^2 + (B_e + c_d B_c r_*^d)L_*. \tag{51}$$

The parameters K, B_c, B_e are nonnegative. Here K bounds the pressure Hessian above, while B_c and B_e bound the negative part of the initial symmetric gradient inside and outside the fixed ball of radius $a_d r_*/\ell_*$, respectively. The boundary penalty, denoted $\mathcal{L}$ to distinguish it from L_*, is chosen at least $C_{1,d}B_c$.

Lemma 7.1 (Fixed-cutoff propagation)**.** *For a constructed child let* $e_p \geq 0$ *be its upper pressure-Hessian increment and* $e_i \geq 0$ *its uniform initial-gradient increment. Retain the same physical* ℓ_*, r_*, *replace*

$$K \mapsto K + e_p, \quad B_e \mapsto B_e + e_i, \quad B_c \mapsto B_c + e_i, \quad \mathcal{L} \mapsto C_{1,d}(B_c + e_i).$$

All corresponding mean inequalities hold whenever the resulting $\mathcal{M}$ *is less than* $1/2$, *and*

$$\mathcal{M}_{\mathrm{child}} - \mathcal{M}_{\mathrm{parent}} = \tfrac{1}{2}e_p L_*^2 + (1 + c_d r_*^d)e_i L_*. \tag{52}$$

Proof. The pressure upper bound adds e_p. If the difference of the initial gradients has operator norm at most e_i, its quadratic form is at least $-e_i|z|^2$. The old interior and exterior inequalities therefore persist on exactly the same regions with the indicated new constants. The new boundary penalty meets the same coercivity requirement. Expanding (51) proves (52). On any shorter horizon the reserve only decreases. The child's flow, inverse, time derivatives and pressure are the actual ones already constructed by the packet solve; changing the parameters for the next mean problem does not replace that child. $\square$

A one-time initialization suffices. If $b = KL_*^2/2 + B_eL_* < 1/2$, put $D = c_dB_cr^dL_*$, $g = 1/2 - b$, and $a = \min(1, g/(2(D+1)))$. Replace ℓ, r by $a\ell, ar$ once. The physical splitting radius is unchanged, while $Da^d \le Da \le g/2$, so the full reserve becomes strictly less than 1/2. This choice uses only the already constructed initial stage. Its resulting parameters are frozen for every later stage.

Let $b_* < 1/2$ bound that initialized reserve, and use the fixed dimensional upper bound $\lambda_d = 1+c_d4^{-d}$ for $1+c_dr_*^d$. In the common scale choice, give the primary terms of $\lambda_d(e_pL_*^2/2+e_iL_*)$ total allowance at most $(1/2-b_*)/4$, and give their frequency remainders the same total allowance. The stage estimates split the two errors into these primary terms and remainders. Summing (52) over any constructed finite prefix then yields

$$\mathcal{M}_N \le \frac{b_* + 1/2}{2} < \frac{1}{2}.$$

The multiplier λ_d depends only on d and is included in the finite list of construction constants before choosing the starting index and seed value. The coefficient estimates in Section 8 verify the required bounds at the prescribed frequencies.

Lemma 7.2 (One compact set for all initial increments)**.** *With this fixed-cutoff construction every initial increment is supported in $\overline{B}(0, \max(1/4, \ell_*^{-1}))$, irrespective of its frequency, localization length, or finite truncation order.*

Proof. At a stage with physical localization length $0 < \rho \le 1$, normalized labels are $y = x/\rho$, and the mean cutoff is $\chi((\rho\ell_*)y)$. The actual mean boundary condition expresses its initial velocity as the compactly supported solenoidal boundary operator with this cutoff, multiplied by its scalar penalty. Hence the normalized mean initial field vanishes for $|y| > (\rho\ell_*)^{-1}$. Physical unscaling replaces y by x/ρ, so it vanishes for $|x| > \ell_*^{-1}$. This is true of every actual mean grade. The oscillatory profiles and their full tensor correctors are supported in $|y| \le 1/4$, hence after unscaling in $|x| \le \rho/4 \le 1/4$. The exact residual-removing correction has zero initial value. Summing the actual finite family, including its terminal corrector, proves the assertion. Adding the fixed compact support of the base datum gives one compact set for all finite-stage initial velocities and their pointwise limit. $\square$

8 Joined source estimates and a uniform parameter envelope

We fix the dimension throughout this section. The Sobolev base order $q = 2d+5$ is fixed before any stage or truncation is chosen. Increasing this order changes constants and polynomial degrees only.

8.1 The variational history inverse

For every forced high grade, the inverse consists of a Dirichlet problem before activation and a forward evolution after activation. The Dirichlet coercivity estimate gives polynomial dependence on the parent coefficients on the history interval. On the forward interval this dependence follows from the relative propagator estimate. Compact-interval existence of a forward fundamental matrix alone does not provide either quantitative bound.

The quantitative estimate is established in the transverse space $U = m_0^{\perp}$, of dimension $d-1$. Let

$$Q(t,x) = F(t,x)|_U, \qquad Q_t = Q_1, \qquad Q_{tt} = -\mathcal{H}Q, \qquad \|Qv\|^2 \geq c\|v\|^2, \quad c > 0.$$

The symmetric matrix $\mathcal{H}$ is the actual pressure Hessian along the old particle map. For a positive activation time $0 < t_0 \leq T$, suppose on $[0, t_0]$

$$\langle \mathcal{H}v, v\rangle \leq K\|v\|^2, \qquad 0 \leq K, \qquad Kt_0^2/2 \leq 1/2.$$

These are the inherited one-sided pressure bounds. All norms below can also be taken in spatial-angular L^2; the multiplication coefficients have uniform pointwise bounds.

For a forcing f, solve for $z \in H_0^1((0,t_0);U)$:

$$\int_0^{t_0} \langle (Qz)', (Q\zeta)'\rangle - \langle \mathcal{H}Qz, Q\zeta\rangle \, dt = -\int_0^{t_0} \langle f, Q\zeta\rangle \, dt \quad (\zeta \in H_0^1((0,t_0);U)). \tag{53}$$

Its physical Jacobi displacement is $y = Qz$; its physical high velocity is

$$b = Qz', \tag{54}$$

which is generally different from y'. In particular, $z(0) = z(t_0) = 0$ does not assert that $b(0)$ or $b(t_0)$ vanishes.

Lemma 8.1 (Polynomial history inverse)**.** *The solution of* (53)*, its high velocity, its time derivative and its terminal high velocity have bounds polynomial in*

$$1 + t_0 + t_0^{-1} + c^{-1} + \|Q\|_\infty + \|Q_1\|_\infty + \|\mathcal{H}\|_\infty.$$

Their fixed-order spatial-angular Sobolev bounds are also polynomial in the corresponding fixed-order coefficient bounds. The assertion holds in every finite transverse dimension.

Proof. Put $C_0 = \|Q\|_\infty$, $C_1 = \|Q_1\|_\infty$ and $L = (Q^*Q)^{-1}Q^*$. Then $LQ = I$ and

$$\|L\| \leq c^{-1}C_0, \qquad \|L'\| \leq c^{-1}C_1 + 2c^{-2}C_0^2C_1.$$

Consequently, with

$$A = 1 + c^{-1}C_0 + t_0(c^{-1}C_1 + 2c^{-2}C_0^2C_1),$$

the identity $z = Ly$ and the elementary inequality $\|y\|_{L^2} \leq t_0\|y'\|_{L^2}$ give $\|z'\|_{L^2} \leq A\|y'\|_{L^2}$. The sharper primitive bound $\|y\|_{L^2}^2 \leq (t_0^2/2)\|y'\|_{L^2}^2$ gives

$$\int_0^{t_0} (|y'|^2 - \langle \mathcal{H}y, y\rangle) \geq \tfrac{1}{2}\|y'\|_{L^2}^2 \geq (2A^2)^{-1}\|z'\|_{L^2}^2.$$

Thus the coercive theorem on the fixed Hilbert space H_0^1 constructs the inverse. Testing with z also gives

$$\|y'\|_{L^2} \leq 2t_0\|f\|_{L^2}, \qquad \|z'\|_{L^2} \leq 2At_0\|f\|_{L^2}.$$

Both bounds are polynomial in C_0, C_1 and $\|\mathcal{H}\|$.

Integration by parts and $Q'' = -\mathcal{H}Q$ give

$$Q^*(Qz'' + 2Q_1z') = Q^*f.$$

Therefore the actual acceleration is

$$z'' = (Q^*Q)^{-1}\{Q^*f - 2Q^*Q_1z'\}, \tag{55}$$

and

$$\|z''\|_{L^2} \le c^{-1}C_0(\|f\|_{L^2} + 2C_1\|z'\|_{L^2}).$$

The time trace inequality

$$\|z'\|_{C_t} \le t_0^{-1/2}\|z'\|_{L^2} + t_0^{1/2}\|z''\|_{L^2}$$

bounds both endpoints polynomially. For continuous f, (55) gives continuous z'', and $b' = Q_1z' + Qz''$ is the actual derivative on the closed interval. In particular the forward initial datum $b(t_0)$ is a bounded trace of the solution of (53).

Spatial derivatives are obtained by differentiating the operator on the fixed space $H_0^1((0,t_0);U)$. Every commutator contains only coefficient derivatives and a lower derivative of the same solution. Applying the same coercive inverse proves the fixed-order estimates by finite induction. For example, a sufficient inverse cost is

$$I_0 = I, \qquad I_{r+1} = I + I_r + 2^rBI_r^2,$$

where I bounds the base inverse and B the coefficient block. This is a polynomial after the fixed number q of steps. The Gram inverse in (55) has the same positive-order commutator proof. These arguments use adjoints, norms and Hilbert-space coercivity, and contain no three-dimensional operation. □

On $[t_0, T]$ solve the actual forward transverse equation with initial datum $b(t_0)$ obtained above. Its value and derivative match those of the history, by the same equation and forcing at the junction. The algebraic normal pressure is reconstructed from this joined field. The profile is $\gamma(t) = \alpha$ on the history interval and $\gamma(t) = \alpha g(t)$ afterwards, where $g(t_0) = 1$ and $\alpha > 0$. Normalization by γ in spatial word estimates never differentiates γ in time.

8.2 The full transverse relative estimate

We give the additional-dimensional calculation explicitly. At activation let p, q be the old unit normal and unit velocity, and put

$$a = \langle Bq, p\rangle > 0, \qquad \beta = \frac{\|\Pi_{\{p,q\}^\perp}Bq\|}{a}, \qquad \sigma = \sqrt{\beta}, \qquad \varepsilon = \sqrt{a/h}.$$

Here h is the old shear height. We use $0 < \sigma \le 1/4$, $1/2 \le a \le 2$ and $0 < \varepsilon \le 1$. Choose the third unit vector n in the direction of $\Pi_{\{p,q\}^\perp}Bq$; the positive tilt makes it nonzero. Write $E = \operatorname{span}\{p, q, n\}$ and $\mathcal{E} = E^\perp$. The center reflection fixes E and reverses $\mathcal{E}$. The actual center coefficient therefore preserves this splitting. Neighboring coefficients need not preserve it.

Use the inherited bounds $G \ge 1$, $\|B\| \le G$ and $\|B_t\| \le G^2$. Transport the entire orthonormal frame by its skew connection S. In this frame, for the old background matrix B,

$$S = p \otimes B^*p - B^*p \otimes p + q \otimes \Pi_{\{p,q\}^\perp}Bq - \Pi_{\{p,q\}^\perp}Bq \otimes q.$$

Consequently $\|S\| \le 4G$ if $\|B\| \le G$, and the exact column identity is

$$(B + S)q = 2\langle Bq, p\rangle p + \langle Bq, q\rangle q. \tag{56}$$

The normal-normal connection is zero. Differentiating the transported background gives norm at most $G^2 + 2(4G)G = 9G^2$. At activation its column outside the q diagonal is $a(p + \beta n)$.

At later physical time t its defect is therefore at most $9G^2|t-t_0|$. These statements concern the complete vector column, including all additional components.

Use fast time $\tau = a(t-t_0)/\varepsilon$. Only the actual interval $0 \le \tau \le \tau_{\text{end}}$ is needed; let $\tau_{\text{end}} \le \Theta$, where $\Theta \ge 1$ is the numerical horizon. No actual coefficient is extended to a longer interval. At a neighboring label the actual gradient in this same frame is

$$M = B + h(t)\, q \otimes p + \mathcal{R}, \qquad \|\mathcal{R}\| \le e_{\text{old}} + \rho L_M.$$

The rank-one term uses the center pair; the neighboring discrepancy is included in $\mathcal{R}$.

Define the diagonal maps

$$D_r = \operatorname{diag}(1, \varepsilon, 1, I_{\mathcal{E}}), \qquad D_v = \operatorname{diag}(\varepsilon, 1, \varepsilon, \varepsilon I_{\mathcal{E}}).$$

They satisfy $D_r D_v = \varepsilon I$ and both have norm at most one. The actual ray and high velocity, including every extra coordinate, are

$$m = s_0 O(t) D_r y, \quad y = (P, Q, N, r), \qquad v = O(t) D_v z, \quad z = (U, V, W, Z), \quad r, Z \in \mathcal{E}. \tag{57}$$

The primary uses one fixed coordinate endpoint at every label. More precisely, if the endpoint lemma selects the physical vector $Y_{\text{phys}} \in m(t_0, 0)^\perp$, set

$$Y_{\text{coord}} = R_\perp^* F(t_0, 0)^{-1} Y_{\text{phys}}.$$

At the center, $F(t_0, 0) R_\perp Y_{\text{coord}} = Y_{\text{phys}}$. At every other label the history has the same coordinate endpoint Y_{coord}; no endpoint is selected again. Its norm is at most $\|F(t_0, 0)^{-1}\| \|Y_{\text{phys}}\|$. Thus Y_* in the history Lipschitz estimates below includes this old inverse-frame factor.

For $t_0 > 0$, the unlocalized history is solved pointwise in the finite-dimensional transverse space and has uniform spatial derivative bounds. Its cylinder L^2 realization is obtained after multiplication by the compact seed. To verify this directly, put $Z(y, \theta) = \chi_{\text{in}}(y) f_\delta(\theta) Y_{\text{coord}}$ and $z_{\text{lin}}(t, y, \theta) = t Z(y, \theta)/t_0$. The zero-trace remainder $w = z - z_{\text{lin}}$ solves (53) with forcing

$$f_{\text{lin}} = -(2/t_0) Q_1 Z.$$

This identity follows from $Q_{tt} = -\mathcal{H} Q$ and $z_{\text{lin,tt}} = 0$. The forcing is a continuous path in every cylinder Sobolev space, is angular-mean zero, and has compact spatial support. Thus the Hilbert-space inverse applies. Its uniqueness identifies the result with the localized pointwise history. All cylinder estimates concern this localized field. For $t_0 = 0$ the prescribed forward primary is used instead of the history construction.

Here $s_0 > 0$ and $O(t)$ is the actual transported orthogonal frame. The velocity v is the unamplified selected high solution. The packet uses αv, with $\alpha > 0$ chosen from its actual target size below. Set

$$A = a^{-1} D_r M D_v, \qquad C = a^{-1} D_r (M + S) D_v.$$

Direct substitution into the ray and high equations gives

$$y' = (C^* - 2A^*) y, \qquad z' = \mathcal{B}_\varepsilon(y) z, \qquad \mathcal{B}_\varepsilon(y) = -C + 2 \frac{D_r^2 y}{|D_r y|^2} \otimes A^* y. \tag{58}$$

For example, the pressure numerator transforms by $\langle D_r y, M D_v z \rangle = a \langle y, A z \rangle$. Thus (58) is the actual rescaled equation; no extra-mode equation has been suppressed.

Let

$$A_0 = (p + \beta n) \otimes q + q \otimes p, \qquad C_0 = 2p \otimes q + q \otimes p.$$

The only large shear scales to $(\varepsilon^2 h(t)/a) q\otimes p$. For any matrix L, separating its q input column gives

$$a^{-1} D_r L D_v = (\varepsilon/a) D_r L + ((1-\varepsilon)/a)(D_r L q)\otimes q.$$

Together with (56), this proves

$$\max(\|A-A_0\|, \|C-C_0\|) \le \left|\frac{\varepsilon^2 h(t)}{a} - 1\right| + \frac{11\varepsilon G + 18G^2|t-t_0| + e_{\text{old}} + \rho L_M}{a}. \tag{59}$$

One obtains the slightly better coefficient 3 in place of 11 for A; the displayed common bound suffices. The norm of an additional input column has the small factor ε: the shear is zero on $\{p,q\}^\perp$. In particular there is no estimate of an extra column by the full large shear. Nor does matrix scaling introduce ε^{-1}, since both diagonal maps have norm at most one.

The logarithmic derivative of $h(t)$ is bounded by $2G$. For $G\varepsilon\Theta/a$ small, its contribution to (59) is $O(G\varepsilon\Theta/a)$. Choose a single error parameter $e \ge \varepsilon$ large enough that the right side of (59) is at most $2e$, and that the initial ray error is at most e. It is bounded by a fixed constant times

$$\varepsilon\Theta G^2 + e_{\text{old}} + \rho\left(L_M + \frac{3L_m}{s_0\varepsilon} + \frac{2L_{\text{hist}}Y_*}{\varepsilon}\right). \tag{60}$$

The lower bound on a was used here. The constant L_m bounds the spatial Lipschitz norm of the normal at activation. The constant L_{hist} bounds, in operator norm, the spatial derivative of the linear map $Y_{\text{coord}} \mapsto b_{Y_{\text{coord}}}(t_0,y)$. Finally Y_* bounds the norm of the selected coordinate endpoint. Consequently the spatial variation of its high velocity is at most $\rho L_{\text{hist}} Y_*$.

The reference ray is

$$y_0(\tau) = (\beta\tau^2, -2\beta\tau, 1, 0).$$

Its full generator $C_0^* - 2A_0^* = -p\otimes q - 2\beta q\otimes n$ is nilpotent of order three. Its evolution is exactly

$$I - (\tau-s)p\otimes q - 2\beta(\tau-s)q\otimes n + \beta(\tau-s)^2 p\otimes n$$

and has norm at most $5\Theta^2$. Duhamel and a supremum absorption now give, if $60e\Theta^3 \le 1$,

$$\|y-y_0\| \le 310e\Theta^5. \tag{61}$$

Indeed the difference generator has norm at most $6e$; the resulting bound is $10\Theta^2(e+30e\Theta^3) \le 310e\Theta^5$. Under the stronger smallness condition used below, this is at most $1/2$. Thus

$$N \ge \tfrac{1}{2}, \qquad |D_r y|^2 \ge \tfrac{1}{4}, \qquad \|y\| \le 6\Theta^2.$$

In particular the extra ray components r are controlled by (61); they have not been set to zero.

Tangency is $y\cdot z = 0$, so the eliminated coordinate is exactly

$$W = -\frac{PU + QV + r\cdot Z}{N}. \tag{62}$$

Let $\zeta = (U,V,Z)$, let Π omit the n coordinate, and let $E_y\zeta$ insert the value (62). Then $\|E_y\| \le 1 + 2\|y\| \le 13\Theta^2$, and the actual reduced equation on the full $d-1$ dimensional transverse space is $\zeta' = \Pi\mathcal{B}_\varepsilon(y)E_y\zeta$. It retains all couplings to Z.

The pressure denominator admits the following operator-norm estimate on the full ambient space. If $r_* = \|y-y_0\| \le 1/2$, $R_* \ge \max(\|y\|, \|y_0\|)$ and $\eta = \max(\|A-A_0\|, \|C-C_0\|) \le 1$, then

$$\begin{aligned}\|\mathcal{B}_\varepsilon(y) - \mathcal{B}_0(y_0)\| \le \eta &+ 8R_*(4+8R_*^2)(r_* + \varepsilon R_*)\\ &+ 2R_*(\eta R_* + 3r_*).\end{aligned} \tag{63}$$

To prove it, put $u_\varepsilon = D_r^2 y/|D_r y|^2$. The preceding lower denominator bounds and $\|D_r y - D_0 y_0\| \le r_* + \varepsilon R_*$ give $\|u_\varepsilon - u_0\| \le (4 + 8R_*^2)(r_* + \varepsilon R_*)$. Also $\|A^* y - A_0^* y_0\| \le \eta R_* + 3r_*$, $\|A^* y\| \le 4R_*$ and $\|u_0\| \le R_*$. Insert these estimates in the difference of the two rank-one operators in (58). All estimates are Euclidean operator estimates on the complete space.

Using (61), $\eta \le 2e$, $R_* \le 6\Theta^2$, and $\varepsilon \le e$, the right side of (63) is at most $5 \cdot 10^6 e\Theta^{11}$. The ideal generator has zero n input column. Therefore $\Pi\mathcal{B}_0(y_0)(E_y - E_{y_0}) = 0$, and the actual reduced generator differs from the ideal reduced generator by at most

$$65 \cdot 10^6 e\Theta^{13}. \tag{64}$$

The ideal reduced generator is explicitly

$$\begin{pmatrix} \dfrac{2P_0Q_0}{1+P_0^2} & -2 + \dfrac{2P_0(P_0+\beta)}{1+P_0^2} \\ -1 & 0 \end{pmatrix} \oplus 0_{\mathcal{E}}, \qquad P_0 = \beta\tau^2, \quad Q_0 = -2\beta\tau.$$

Hence its active equation is

$$((1+\beta^2\tau^4)V')' = 2(1-\beta^2\tau^2)V, \qquad U = -V',$$

and every ideal extra coordinate is constant.

Let $g = V_0$ be the positive scalar solution with $g(0) = 1$, $g'(0) = 0$. The scalar relative propagator and the ratio estimate $g(s)/g(t) \le C\Theta$ bound the complete ideal reduced evolution by

$$K_d \frac{g(t)}{g(s)}, \qquad K_d = C_d\Theta^8.$$

The extra identity block is included using this ratio estimate; g need not be nondecreasing. For completeness, whenever the reduced coefficient gap is Δ, weighted Duhamel gives

$$\frac{g(s)}{g(t)}\|U(t,s)\| \le K_d + K_d\Delta \int_s^t \frac{g(s)}{g(r)}\|U(r,s)\|\,dr.$$

Thus $K_d\Theta\Delta \le 1/2$ implies $\|U(t,s)\| \le 2K_d g(t)/g(s)$. This estimate covers arbitrary transverse initial data and forcing, including additional modes.

It also bounds the true joined high evolution in the fixed transverse coordinates used in the history problem. At each endpoint the actual physical field and reduced coordinates are related by

$$b = OD_vE_y\zeta, \qquad \zeta = \Pi D_v^{-1}O^*b.$$

Their operator norms are at most $13\Theta^2$ and ε^{-1}, respectively. If $R_\perp : U \to m_0^\perp$ is an isometric identification and $b = FR_\perp a_{\text{hist}}$, then

$$a_{\text{hist}} = R_\perp^* F^{-1} b, \qquad b = FR_\perp a_{\text{hist}}.$$

These two endpoint changes cost at most $\|F^{-1}\|$ and $\|F\|$. Thus the evolution in the actual a_{hist} coordinates is bounded by $26K_d\Theta^2\varepsilon^{-1}\sup\|F^{-1}\|\sup\|F\|\,g(t)/g(s)$. Every factor is a known polynomial source cost. Applying the same endpoint maps to the variation-of-constants formula gives the forced tail bound with these costs. This is the full $d-1$ dimensional forward operator joined to the history trace, including its action on all additional transverse components.

For the selected primary let $\lambda \ge 0$ be its actual normalized initial slope and let $V_\lambda(0) = 1$, $V_\lambda'(0) = \lambda$. The ideal reduced initial vector is $(-\lambda, 1, 0)$. At a neighboring label the discrepancy of (U, V, Z) is at most $\rho L_{\text{hist}} Y_*/\varepsilon$, hence at most e after the choice (60). In particular the extra primary initial component is bounded by the same history Lipschitz estimate. The remaining initial component is determined by (62).

Weighted Duhamel applied to the difference from $\zeta_\lambda = (-V_\lambda', V_\lambda, 0)$ now yields

$$\|\zeta - \zeta_\lambda\| \le C_d e \Theta^{30}(1+\lambda) g(\tau). \tag{65}$$

Indeed the initial term is $K_d e\, g(\tau)$ and the integral term is at most $2K_d^2 \Theta\Delta(1+\lambda+e)g(\tau)$. The exponent is $2\cdot 8 + 1 + 13 = 30$. Restoring the eliminated coordinate costs at most two more powers of Θ, by (62) and (61).

There is no loss of λ on the late interval. The second fundamental solution satisfies

$$\frac{V_1(\tau) - g(\tau)}{g(\tau)} = \int_0^\tau \frac{ds}{(1+\beta^2 s^4) g(s)^2}.$$

On $[0,1]$ the scalar equation has uniformly bounded coefficients, so this integral at one has a fixed positive lower bound. Consequently, for $\tau \ge 1$, $V_\lambda(\tau) \ge c(1+\lambda)g(\tau)$. The actual normalized flux is $J = \langle y, Az\rangle$; the physical flux for the unamplified field equals $s_0 a J$; the amplified field has flux $\alpha s_0 a J$. The ideal flux is $J_0 = Q_0(-V_\lambda') + (P_0+\beta)V_\lambda$. Equations (61)–(65) give

$$|J - J_0|/V_\lambda \le C_d e \Theta^{34} \quad (1 \le \tau \le \tau_{\mathrm{end}}). \tag{66}$$

For example, expand its difference into $\langle y, (A-A_0)z\rangle + \langle y - y_0, A_0 z\rangle + \langle y_0, A_0(z - z_\lambda)\rangle$. The final term costs two powers of Θ beyond the restored-state bound; the other two terms are smaller.

The scalar formula gives $J_0/V_\lambda \ge \beta = \sigma^2$. To recall its margin, below $x = \sigma\tau = 1$ both the $Q_0(-V_\lambda'/V_\lambda)$ contribution and P_0 are nonnegative. Above $x=1$ its equivalent expression is $x^2 - \beta + 2\sqrt{\beta}\, z/x \ge 1 - \beta \ge 3/4$, with the nonnegative scalar Riccati variable z. Let $x_{\mathrm{tar}} \ge 2$ be the target coordinate and assume $\tau_{\mathrm{tar}} = x_{\mathrm{tar}}/\sigma \le \Theta$. Then $\beta \ge 4/\Theta^2$. A single conservative guard

$$C_d e \Theta^{60} \le 1 \tag{67}$$

therefore proves positive actual late flux, as well as all the ray, denominator, reduced-propagator and relative-state guards above. Enlarge the fixed C_d to cover their finitely many numerical constants.

The same guard transfers the ideal uniform-size estimate to the actual neighbor. Indeed the unamplified size is $s_0\|D_r y\|\|D_v z\|$; the packet size is α times this quantity. Subtract the corresponding central/reference expression for the packet size and use the two state bounds. At the target the actual q velocity component is at least $V_\lambda(\tau_{\mathrm{tar}})/2$ and the normal component N is at least $1/2$. Hence the unamplified target size is at least $s_0 V_\lambda(\tau_{\mathrm{tar}})/4$. The actual normalization $\alpha\, \mathrm{size}_{\mathrm{target}} = \delta h_{\mathrm{child}}$ and the scalar ratio $g(\tau)/g(\tau_{\mathrm{tar}}) \le C\Theta$ make this size error at most $C_d e \Theta^{40} \delta h_{\mathrm{child}}$, hence at most $\delta h_{\mathrm{child}}$. On $0 \le \tau \le 1$, the reference states are at most $C(1+\lambda)$ and the perturbation bound adds at most the same size. The scalar target growth gives $\alpha s_0(1+\lambda) \le C\Theta\delta h_{\mathrm{child}} e^{-b/\sigma}$. Here $b > 0$ is a fixed constant from the scalar growth lemma, decreased if needed. Thus the early size retains its exponentially small factor and a fixed polynomial in Θ (the larger Θ^5 bound suffices). All these statements concern the actual short interval ending at τ_{end}; the ideal scalar equation alone is defined globally.

The bounded profile height. The preceding normalization also closes the polynomial bound for the profile constant H_0 through cancellation of the scalar growth against the terminal normalization. The forward scalar profile is $g = V_0$. In physical time this means $\gamma(t) = \alpha$ for $t \le t_0$ and $\gamma(t) = \alpha V_0(a(t-t_0)/\varepsilon)$ for $t \ge t_0$. Since $V_0 \le V_\lambda$, the scalar ratio estimate on the retained interval through the target and its stipulated short extension gives $g(\tau)/V_\lambda(\tau_{\mathrm{tar}}) \le C\Theta$. Together with the actual target-size lower bound this yields

$$H_0 = \max(1, \sup\gamma) \le 1 + C\Theta\, \delta h_{\mathrm{child}} s_0^{-1}. \tag{68}$$

The constant s_0 is that of the actual unit initial phase: $s_0^{-1} = \|F(t_0)^* n_{\mathrm{new}}\| \le \|F(t_0)\|$. It is therefore an old particle-label cost. Likewise the scalar target lower bound gives

$$\alpha \le C\Theta\, \delta h_{\mathrm{child}} s_0^{-1}(1+\lambda)^{-1} e^{-b/\sigma}.$$

Thus both the polynomial height and the small history amplitude come from normalization at the same target; the resulting source majorant has polynomial dependence on the parent parameters.

Finally, (60) explains how to choose a quantitative spatial neighborhood. Only old unnormalized coefficient and history Lipschitz costs occur. Their positive-word inverse and endpoint estimates are polynomial in the old particle-label data. These costs depend only on the parent geometry, its selected history and ε^{-1}. In particular, they are independent of the new angular width δ_n. The prescribed ρ_n below satisfies the resulting neighborhood conditions.

8.3 One radius and polynomial source costs

For an ordered-word Gevrey block put $\Updownarrow_{R,b}(n) = R^{n+b}((n+b)!)^2$. The coefficient fixed-base cost is bounded by the explicit polynomial

$$S_{d,q}(R_c, C) = 2^q C \sum_{r=0}^{q} (4(d+1)R_c)^r (r!)^2.$$

Here R_c is the known coefficient radius. The full word alphabet is $\{1, \dots, d+1\}$.

Indeed, at external order n and base order $0 \le r \le q$,

$$(n+r)!^2 \le 4^{n+r}(n!)^2(r!)^2.$$

The $(d+1)^{n+r}$ ordered coordinate choices therefore give a coefficient-block bound

$$S_{d,q}(R_c, C)\,[4(d+1)R_c]^n (n!)^2.$$

Thus the known coefficient radius is replaced once by $4(d+1)R_c$. In the commutator argument below R_c denotes this enlarged known radius. This conversion is applied only to coefficient tensor bounds; the forcing is already measured by its full ordered-word block.

For a positive continuous profile γ and a fixed base order q, the normalized block used here is

$$\mathcal{B}_n^{\gamma}(f) = \sum_{I \in \{1,\dots,d+1\}^n} \|\gamma^{-1}\partial_I f\|_{C([0,T];H^q)}.$$

The finite sum is outside the time norm. Replacing it by a supremum of the sum is unnecessary in the induction. At each fixed time the block dominates the corresponding sum of Sobolev norms and hence the weighted norms used in the correction estimates. Equivalent choices of the base H^q norm change only constants depending on d, q.

Lemma 8.2 (Same-radius joined bounds)**.** *There is a threshold $R_* \ge 1$, polynomial in the known source coefficients, their reciprocal coercivity bounds, inverse time lengths and the relative propagator constant of Section 8.2, such that for every $R \ge R_*$ the joined high inverse maps a forcing satisfying $\mathcal{B}_n^{\gamma}(f) \le A\Updownarrow_{R,b}(n)$ to velocity and time derivative bounds*

$$\mathcal{B}_n^{\gamma}(h_f) \le C_{\mathrm{v}} A \Updownarrow_{R,b+3}(n), \qquad \mathcal{B}_n^{\gamma}(\partial_t h_f) \le C_{\mathrm{t}} A \Updownarrow_{R,b+3}(n).$$

Here h_f denotes the joined high field. The constants and radius threshold are independent of A, b and $R \ge R_*$. The mean inverse, pressure inverse and tensor corrector have analogous fixed-shift bounds at this same radius.

Proof. The history base estimate is Lemma 8.1. Its terminal trace supplies the actual forward datum. The forward base estimate is the relative estimate in Section 8.2, on the fixed packet support. Products with the known frame turn coordinate velocities into physical velocities, and the Gram equation controls the true time derivative.

Here is the all-order induction, including the radius issue. Differentiating the actual equation produces positive-order coefficient commutators. The elementary inequality

$$\binom{n}{j}(j!)^2((n-j+b)!)^2 \le ((n+b)!)^2$$

holds for every $b \ge 0$. If M is the fixed-base inverse or forward cost and $BR_c^j(j!)^2$ is the known coefficient block, the normalized positive-order sum is bounded by

$$MB\sum_{j\ge 1}(R_c/R)^j \le \tfrac{1}{2} \quad \text{when} \quad R \ge R_c(1+2MB).$$

Thus a factor two closes the induction at all orders, without changing the unknown forcing radius. The subsets of an ordered word give exactly the binomial convolution above. A dimensional factor is permitted when converting the *known* coefficient tensor bounds into word bounds; it is absorbed once into R_c.

The fixed-base inverse is the finite polynomial recursion in Lemma 8.1. The mean form and pressure equation have the same commutator identity and use their actual coercive inverses. The corrector uses known coefficient products, one angular primitive, and one spatial derivative. The primitive costs at most the fixed period and the coordinate sums are finite ($\le d^3$ terms). Taking the sum of these finitely many required radii gives a polynomial common radius. Linearity, applied after dividing the forcing and datum by A, proves that the radius is independent of A; the zero-amplitude case is uniqueness. The time profile itself is never differentiated. □

These estimates define an explicit polynomial source majorant. Replace known primitive bounds by $W \ge 1$, coefficient blocks by $S_{d,q}$, fixed inverses by their q-step polynomial recursion, and maxima by sums. Reciprocal coercivity, ray and Gram constants are first replaced by their *proved* bounds. Every remaining operation in the initialized source construction is addition, multiplication, a fixed power or one of these fixed recursions. The resulting positive polynomial $\Phi_d(W)$ gives the bound

$$\Phi_d(W) \le A_d W^{e_d}, \qquad e_d = \deg \Phi_d, \qquad A_d = 1 + \sum_j |[W^j]\Phi_d|.$$

The exponent is therefore determined by a fixed finite computation. The mean and high grade recursion uses this common radius at every grade.

8.4 The explicit nonlinear threshold

Let $p \ge 300$ be a fixed tail-polynomial degree and $\vartheta = 1/(1000(p+1))$. The grade bounds in the finite recursion permit $p = 300$. Write $Z = k^{\vartheta}$, $N = \lfloor Z \rfloor$. Let $C, I, G, D, T, r_{\mathrm{a}}^{-1}$ be the actual tail, inverse, correction-growth, drift, time and reciprocal radius costs of that packet, with $C, I, G, D, T \ge 0$. Here $r_{\mathrm{a}} > 0$ is its analytic radius; it is distinct from the physical localization length ρ and from the fixed mean cutoff parameter ℓ_*. Define

$$H = 64 + \exp(1) + 2^p C + I + 12GT + \frac{8GTD}{r_{\mathrm{a}}} + \left(\frac{8GT}{r_{\mathrm{a}}}\right)^2, \qquad k_* = H^{1000(p+1)}. \tag{69}$$

For $k \ge k_*$ one has $H \le Z$, $Z \ge 64$ and $\log k \ge 1$. In particular

$$C(N+1)^p \le C(2Z)^p \le Z^{p+1} = k^{1/1000}.$$

The genuine residual and correction tolerance obey

$$2[2e^{-(7/10)Z\log k}]e^{3GT} \le \tfrac{1}{2}e^{-\sqrt{Z}}, \qquad 2G(D/k + e^{-\sqrt{Z}})T \le r_{\mathrm{a}}/2.$$

Indeed $3GT \le Z/4$, $\sqrt{Z} \le Z/8$, $\log 8 \le Z/8$ prove the first inequality. For the second, use $8GTD \le r_{\mathrm{a}}k$ and $8GT/r_{\mathrm{a}} \le \sqrt{Z} \le e^{\sqrt{Z}}$. This gives a polynomial frequency threshold for the exponential correction-growth condition.

If the six costs are bounded by $Q \ge 1$, the explicit bound is

$$k_* \le (2^p + 200)^{1000(p+1)} Q^{6000(p+1)}. \tag{70}$$

For $p = 300$ the power of Q is $1\,806\,000$. Its size is irrelevant; its independence of the stage and the truncation is essential.

8.5 The actual next particle-label bound

Here u denotes the coordinate packet in normalized parent labels, and m_0 is the constant initial covector. The corrected packet's lifted velocity is

$$\mathcal{V} = (k^{-1}u, \langle m_0, u\rangle), \qquad \|m_0\| = 1.$$

Its normal cancellation and the drift energy bound give supremum and cylinder L^2 jet amplitudes

$$B \le C_d(D/k + C_{\mathrm{e}}e^{-\sqrt{k^{\vartheta}}}), \qquad R_* \le C_d/r_{\mathrm{a}}.$$

The actual time derivative has amplitude C_1, a polynomial source cost, at a fixed enlargement of R_*. Choose a fixed $\chi < \min(1/100, \vartheta/100)$. The source comparison below yields $C_dD, C_dC_{\mathrm{e}}, R_*, T, C_1 \le k^{\chi}$. A fixed numerical lower bound on k then gives

$$B \le 2k^{-1/2}, \qquad BR_*T \le 1/8.$$

For example, putting $a = \vartheta/2$ and $v = \log k$, the inequality $e^{av} \ge a^2v^2/2$ shows that $v \ge 2(2+\chi)/a^2$ implies $k^{\chi}e^{-\sqrt{k^{\vartheta}}} \le k^{-2}$.

Lemmas 4.1 and 4.2, the one-dimensional phase trace, and the affine graph derivative factor $1+k$ now give the actual relative-label displacement, velocity and acceleration (a, b, c):

$$\|D^r a\|_{2,\infty},\ \|D^r b\|_{2,\infty} \le k^{-1/4}(\rho^{-1}k^{5/4})^r (r!)^2, \qquad \|D^r c\|_2 \le k^{1/4}(\rho^{-1}k^{5/4})^r (r!)^2.$$

The time acceleration is the actual field $(\mathcal{V}_t + D\mathcal{V}\,\mathcal{V}) \circ \Psi$. The only small-time flow condition is $BR_*T \le 1/8$ on the lifted field.

To record the polynomial degree, let $X = \mathrm{id} + D_0$ be the old particle map, with actual velocity V_0 and acceleration W_0, and $Y = \mathrm{id} + a$ the relative map. Then

$$\begin{aligned}
D_0^+ &= D_0 \circ Y + a,\\
V_0^+ &= V_0 \circ Y + b + (DD_0 \circ Y)b,\\
W_0^+ &= W_0 \circ Y + 2(DV_0 \circ Y)b + (D^2D_0 \circ Y)[b,b] + c + (DD_0 \circ Y)c.
\end{aligned}$$

Suppose the old particle fields have common Gevrey amplitude and radius at most $K \ge 1$, with displacement and velocity controlled in both L^2 and supremum norm and acceleration controlled in L^2. A sufficient new amplitude and radius, with $A = k^{1/4}$ and $R = \rho^{-1}k^{5/4}$, are

$$A^+ = K + A + 9C_dK^2A + 36C_dK^3A^2, \qquad R^+ = (1+R)((1+A)16K + 2) + R.$$

These follow term by term from the factorial product and composition inequalities. Volume preservation permits composition in L^2. If $K, C_d, \rho^{-1} \le k^{1/100}$, fixed numerical margins give

$A^+ \le k$, $R^+ \le k^2$. A fixed Sobolev shift gives the next particle-label bound $K^+ \le k^{2q+4}$. In particular the acceleration has supremum jet bounds after that shift, by the fixed-dimensional Sobolev embedding. This supplies the coefficient bounds for F_2^+ used in the next source.

The full inverse deformation is polynomial as well. Since $\det F^+ = 1$, its inverse is the adjugate. If $\|D^n F^+\| \le AR^n (n!)^2$, expansion of each cofactor into $(d-1)!$ products and the Gevrey product constant 3 give

$$\|D^n(F^{+,-1})\| \le d(d-1)!\, 3^{d-2} A^{d-1} R^n (n!)^2.$$

The factor d bounds operator norm by the largest entry. The time derivative satisfies $(F^{+,-1})_t = -F^{+,-1} F_1^+ F^{+,-1}$. Also $F_1^+ = D_x V_0^+$ and $F_2^+ = D_x W_0^+$. Thus the source coefficients have fixed polynomial bounds in the new particle-label parameter. Fix, for instance,

$$D_d = 100(q + d + 1)$$

for the particle-label invariant $K^+ \le k^{D_d}$. The inverse and coefficient costs are fixed polynomials in K^+; they are not asserted to have that same exponent D_d.

8.6 Uniform scales, with no second frequency choice

Set $s_n = J + n$ and

$$x_0 = X, \quad x_{n+1} = s_n^2 x_n, \quad k_n = e^{x_n/s_n^2}, \quad h_n = e^{x_n/s_n^5}, \quad \delta_n = e^{-x_n/s_n^3}, \quad \rho_n = e^{-x_n/s_n^{7/2}}.$$

Here $n = j - J$ and $x_n = x_{j-1}$ in the notation of Section 10; ρ_n is the physical localization, not the fixed mean cutoff ℓ_*. For $n \ge 1$ the identities

$$\log k_{n-1} = \frac{x_n}{(s_n-1)^4}, \quad \log h_{n-1} = \frac{x_n}{(s_n-1)^7}, \quad \log \rho_{n-1}^{-1} = \frac{x_n}{(s_n-1)^{11/2}}$$

are exact. If $K_n \le k_{n-1}^{D_d}$, every old geometric cost in (60) is bounded by $C s_n^a x_n^b e^{c x_n/(s_n-1)^4}$. For sufficiently large fixed J and then X,

$$\rho_n C s_n^a x_n^b e^{c x_n/(s_n-1)^4} \le e^{-x_n/(2 s_n^{7/2})}.$$

The additional factors in the ambient guard have the same inherited scale: $\Theta_n \le C(1 + s_n^2 x_n^2)$, $s_0^{-1} \le \|F(t_0, 0)\|$, and $\varepsilon^{-1} \le C\sqrt{h_{n-1}}$. They therefore contribute only polynomial factors in s_n, x_n or an exponential with denominator $(s_n - 1)^7$, which is already absorbed by the displayed denominator $(s_n - 1)^4$. The terms $\varepsilon \Theta G^2$ and e_{old} in (60) are controlled by the separate shear separation and old frequency-error comparisons in Section 10; they are not made small by localization. At $n = 0$ the activation time is zero, and the forward high construction is used. This branch contains no inverse-history-time cost. The seed predecessor has polynomial costs in X. The normal-stage factor $e^{-X/J^{7/2}}$ dominates those costs for the same final large choice of X.

Consequently all neighborhood conditions hold at the prescribed localization length ρ_n.

The full source list additionally includes δ_n^{-1} and the new scalar height. Its positive-polynomial bounds therefore have the form

$$\mathcal{P}_n = C s_n^a x_n^b \exp\left(\frac{c x_n}{(s_n-1)^3}\right). \tag{71}$$

In particular the radius, H_0, reciprocal analytic radii, history and mean inverse constants, and the costs in (69) are bounded by fixed powers of (71).

Lemma 8.3 (Uniform domination). *For a finite list of fixed $A, C, c, a, b, N > 0$ and $\gamma > 0$, one common choice of J and then X ensures*

$$A\mathcal{P}_n^N \le k_n^\gamma \qquad (n \ge 0)$$

for every entry of the list.

Proof. Choose J so that $Nc/(s-1)^3 \le \gamma/(2s^2)$ for $s \ge J$; $J \ge 16Nc/\gamma + 2$ suffices. Increase X until

$$\log(AC^N) + Na\log s_n + Nb\log x_n \le \frac{\gamma x_n}{2s_n^2}$$

uniformly in n. To see the uniformity, use $x_n = X\prod_{r<n}(J+r)^2$. The sequences $s_n^2 \log s_n / \prod_{r<n}(J+r)^2$ and $s_n^2 \log(\prod_{r<n}(J+r)^2)/\prod_{r<n}(J+r)^2$ are bounded, since the denominator eventually dominates every polynomial in n while its logarithm grows at most $2n\log(J+n)$. The resulting bound is $O_J((1+\log X)/X)$ and tends to zero. Adding the two inequalities proves the assertion. Take the maximum of the finitely many choices of J, X. $\square$

Apply this lemma simultaneously to the source costs, the neighborhood guards, (70), the small lifted-flow costs and their numerical margins. It gives the actual nonlinear correction at the preselected k_n and then $K_{n+1} \le k_n^{D_d}$. The exponent D_d is fixed first; the source polynomials and their degrees are then fixed; J, X are chosen once for the resulting finite list. No exponent or starting index is enlarged after a later child is observed.

The induction applies to the joined variational-history family and the quantitative neighborhood constructed in this section.

9 Initial increments at every Sobolev order

We use the joined displacement-history and forward inverse of Lemma 6.2. When the activation time is positive, the forced high coefficients generally have nonzero initial velocities; all of them enter the estimates below.

Fix the dimension d and a legal integer base Sobolev order s for the packet estimates. Constants below may depend on d and on a fixed derivative order, but the frequency condition will not depend on that derivative order. Write $\mathcal{P} \ge 1$ for a bound on the actual source primitives, including the coefficient and inverse costs, the terminal datum, the reciprocal phase scale, the fixed mean cutoff cost, and the common Gevrey radius. Section 8 bounds these primitives for the constructed sources, uniformly along the prescribed scale sequence.

9.1 The actual three sums at time zero

Put $\kappa = k^{-1}$ and let N be the chosen truncation. Since the exact nonlinear correction has zero initial value and the parent flow is initially the identity, the joined finite construction gives

$$u_{\mathrm{osc},0}(a) = \rho\sum_{p=1}^{N}\left[\kappa^p A_p + \kappa^{p+1}C_p\right](0, a/\rho, km_0\cdot a/\rho), \tag{72}$$

$$u_{\mathrm{mean},0}(a) = \rho\sum_{p=2}^{N}\kappa^p B_p(0, a/\rho), \qquad \Delta u_0 = u_{\mathrm{osc},0} + u_{\mathrm{mean},0}. \tag{73}$$

Here C_p is the full ambient antisymmetric-potential corrector and B_p is independent of the angle. The estimates include every high coefficient, its tensor corrector, and every mean coefficient. Formula (72) extends [1, §3.9, (3.60)] to the full ambient tensor construction.

Let $\alpha > 0$ be the selected primary amplitude. The joined profile is

$$\gamma(t) = \alpha \quad (0 \le t \le t_0), \qquad \gamma(t) = \alpha g(t) \quad (t_0 \le t \le T), \qquad g(t_0) = 1,$$

and put $H_0 = \max(1, \alpha, \sup \alpha g)$. The actual coefficient induction assigns the oscillatory profiles γH_0^{2p-2} and the mean profiles H_0^{2p-2}.

The factor α on the history interval can be checked directly. The primary terminal displacement is α times the normalized compact datum, and the displacement inverse is linear. In every higher zero-mean force, terms containing only means cancel under angle-mean subtraction. Every remaining term contains a previous oscillatory factor or a derivative of one. The term involving the newly solved mean is $-(m \cdot B_p)\partial_\theta A_1$, which also contains this factor. Extra oscillatory factors are bounded by the indicated powers of H_0. The actual history inverse and the angular tensor primitive therefore retain α at time zero. This factor is preserved at each step of the graded construction by linearity of its high solver.

Suppose first that $t_0 > 0$. In fixed transverse coordinates, every forced history solves $\xi(0) = \xi(t_0) = 0$ and has velocity $A = FR_\perp \xi_t$. The primary instead has $\xi(0) = 0$ and $\xi(t_0) = \alpha \chi_{\text{in}}(y) f_\delta(\theta) \xi_T$, where ξ_T is the actual selected terminal coordinate. It is obtained from the physical central endpoint by the actual inverse frame; its norm is included in $\mathcal{P}$. Subtracting the affine extension $(t/t_0)\alpha\chi_{\text{in}} f_\delta \xi_T$ reduces this primary problem to the same zero-endpoint inverse. All spatial and phase words of the forced problem still have zero endpoint values. For the primary, differentiating this explicit affine extension supplies exactly the differentiated boundary datum; the differentiated boundary traces are therefore explicit. The forward solution starts with the actual trace $FR_\perp \xi_t(t_0)$. It agrees there with the history solution, and the common velocity equation then gives equality of the time derivatives at the junction. These are the boundary conditions of Lemma 6.2. When $t_0 = 0$, the primary instead has its prescribed forward datum and the forced high fields have zero initial velocity. The forward estimates give the same bounds below, with no history inverse or reciprocal factor t_0^{-1}. This applies both to the exceptional seed and to the first normal stage $j = J$; positive history begins at $j = J + 1$.

For completeness, the all-word estimate uses this same inverse. After translation in the label and angle, write the equation as $L(a)u(a) = f(a)$. At a fixed legal base Sobolev order let its inverse cost be M, and let the known coefficient derivative costs be $BR_c^j (j!)^2$. Differentiating a word of length n gives the exact Leibniz identity

$$L\partial^\omega u = \partial^\omega f - \sum_{\varnothing \neq S \subset \{1,\dots,n\}} (\partial^{\omega_S} L)\partial^{\omega_{S^c}} u.$$

The homogeneous boundary space is unchanged, so the same inverse applies to each term. In the block norm this yields a binomial convolution. For $b \ge 0$,

$$\binom{n}{j} (j!)^2 ((n - j + b)!)^2 \le ((n + b)!)^2.$$

Consequently a forcing bound $FR^{n+b}((n+b)!)^2$ gives $2MFR^{n+b}((n+b)!)^2$ for the solution as soon as $R \ge R_c(1+2MB)$: the positive-order convolution costs at most $MB\sum_{j\ge1}(R_c/R)^j \le 1/2$. The initial finite base inverse is obtained by the usual finite Sobolev commutator recursion. Its order is fixed by d, not by the packet grade. Time traces follow from the actual H^1 equation and cost a polynomial in the reciprocal history length and known frame bounds. The same estimate on the forward interval uses its proved relative propagator and the actual history trace. This proves preservation of the unknown forcing radius; known coefficient conversion enlarges R_c only once.

The displacement argument uses the whole transverse space of dimension $d-1$. If $R_\perp$ has orthonormal columns, $\eta = FR_\perp\xi$, and $\eta(t_0) = 0$, then

$$\int_0^{t_0} (|\eta_t|^2 - \langle H\eta, \eta\rangle)\, dt \geq \left(1 - \frac{K_+ t_0^2}{2}\right) \int_0^{t_0} |\eta_t|^2\, dt \geq \frac{1}{2}\int_0^{t_0} |\eta_t|^2\, dt.$$

The one-sided Poincaré inequality here follows by integrating $\eta(t) = -\int_t^{t_0} \eta_t(r)\, dr$. Conversion to ξ uses only the known bounds for F^{-1} and its derivative. Thus the inverse cost is polynomial in those bounds, and this step has no three-dimensional cross-product dependence.

9.2 Separating derivative order from expansion order

For a fixed integer $m \geq 0$, the coefficient estimates need derivatives only through $m + s + 1$. Their shift is at most $100p$; replacing this fixed coefficient by a larger dimensional constant makes no change to the argument. For $b = m + s + 1$,

$$(b + 100p)! \leq 2^{b+100p} b!(100p)!, \qquad ((b + 100p)!)^2 \leq C_b(Cp)^{200p}.$$

Also $R^{b+100p} = R^b(R^{100})^p$. Consequently all dependence on m can be put in a prefactor $C_{d,m}\mathcal{P}^{c_{d,m}}$, while the expansion-order bound has the form

$$B_N^{p+1}, \qquad B_N = \mathcal{P}^{c_d}(C_d N)^{L_d}. \tag{74}$$

with one finite exponent L_d independent of m. For the displayed shifts one may take $L_d = 300$ after enlarging constants. Choose $N = \lfloor k^\vartheta \rfloor$ with $L_d\vartheta < 1/100$. A single fixed source-to-frequency guard then ensures $B_N \leq k^{1/100}$.

At time zero the cylinder Sobolev norms of A_p and C_p are bounded by

$$\alpha C_{d,m}\mathcal{P}^{c_{d,m}} B_N^{p+1}.$$

The mean has the same bound without α. For $k \geq 4$,

$$\sum_{p=1}^{N} k^{-p} B_N^{p+1} \leq C, \quad \sum_{p=1}^{N} k^{-p-1} B_N^{p+1} \leq C, \quad \sum_{p=3}^{N} k^{-p} B_N^{p+1} \leq Ck^{-2}.$$

These follow by summing the geometric series with ratio $k^{-99/100}$. In the last sum its first power is $k^{-3+4/100} \leq k^{-2}$. The grade-two mean is estimated separately at its fixed order, giving $C_{d,m}\mathcal{P}^{c_{d,m}} k^{-2}$ as well. This fixed-order separation prevents an artificial loss of a small power of k in the mean estimate.

For $T_k f(y) = f(y, km_0 \cdot y)$ and $|m_0| = 1$, the chain rule and $H^1(\mathbb{T}) \hookrightarrow L^\infty(\mathbb{T})$ give

$$\|T_k f\|_{H^m(\mathbb{R}^d)} \leq C_{d,m}(1 + k)^m \|f\|_{H^{m+1}(\mathbb{R}^d \times \mathbb{T})}.$$

Indeed each spatial derivative is a product of the commuting operators $\partial_{y_i} + km_{0,i}\partial_\theta$. Apply the periodic one-dimensional estimate to each resulting derivative at fixed y, then integrate in y. Exactly one additional angle derivative is needed. For the mean there is no graph loss because the field is angle independent. Finally, physical scaling multiplies the L^2 norm of an order-r derivative by $\rho^{1+d/2-r} \leq \rho^{-m}$ for $r \leq m$ and $0 < \rho \leq 1$. We obtain the actual estimates

$$\|u_{\mathrm{osc},0}\|_{H^m} \leq C_{d,m}\rho^{-m} k^m \alpha \mathcal{P}^{c_{d,m}}, \tag{75}$$

$$\|u_{\mathrm{mean},0}\|_{H^m} \leq C_{d,m}\rho^{-m} k^{-2} \mathcal{P}^{c_{d,m}}. \tag{76}$$

The selected central amplitude satisfies

$$\alpha \le \mathcal{P}^{c_d} e^{-bx}$$

for a fixed $b > 0$, by the actual central target growth, with the normalized terminal and reciprocal phase costs included in $\mathcal{P}$. In the scalar indexing below, $n = j - J$ corresponds to normal stage j of Section 10: $a_n = j$, $x_n = x_{j-1}$, $k_n = k_j$, and $\rho_n = \rho_j$. The inherited geometric parameter is therefore x_n, while the new target is x_{n+1}. The amplitude factor is e^{-bx_n}. The required exponential gain is present for all initial high coefficients, including those produced by positive history intervals.

9.3 One scale sequence gives all orders of summability

Lemma 9.1 (Scalar summability)**.** *Fix $J \ge 2$, $X \ge 1$, and define*

$$a_n = J + n, \quad x_0 = X, \quad x_{n+1} = a_n^2 x_n, \quad k_n = e^{x_n/a_n^2}, \quad \rho_n = e^{-x_n/a_n^{7/2}}.$$

Here ρ_n is the physical packet length; it is distinct from the fixed physical mean-cutoff parameter ℓ_* and from the Gevrey weight radius used in the cylinder estimates. Let

$$\mathcal{P}_n = C a_n^p x_n^q \exp\left(\frac{c x_n}{(a_n - 1)^3}\right)$$

with fixed $C \ge 1$, $c \ge 0$, $p, q \in \mathbb{N}$ and $b > 0$. For every fixed $m, \nu \in \mathbb{N}$, both series

$$\sum_n \rho_n^{-m} \mathcal{P}_n^\nu k_n^m e^{-bx_n}, \qquad \sum_n \rho_n^{-m} \mathcal{P}_n^\nu k_n^{-2}$$

converge.

Proof. We have $x_n \ge X 4^n$ and $\log x_n \le \log X + 2n \log(J + n)$. Therefore, for every fixed $r \ge 0$,

$$\frac{a_n^r (1 + \log a_n + \log x_n)}{x_n} \longrightarrow 0.$$

The logarithm of the first summand, divided by x_n, tends to $-b$: its other exponential terms are m/a_n^2, $m/a_n^{7/2}$ and $c\nu/(a_n - 1)^3$, and its logarithmic terms vanish. Eventually the summand is at most $e^{-bx_n/2}$, a summable sequence.

For the second summand, divide its logarithm by x_n/a_n^2. The result tends to -2, since

$$m a_n^{-3/2} \to 0, \qquad c\nu \frac{a_n^2}{(a_n - 1)^3} \to 0,$$

and the logarithmic terms again vanish. Eventually it is at most $e^{-x_n/a_n^2} \le e^{-n}$, proving convergence. $\square$

The choices of J, X in this lemma are independent of m. Only the starting index of the convergence comparison depends on m; the construction itself uses the same scales for every Sobolev order. The same proof permits a localization scale whose reciprocal is bounded by a polynomial times $\exp(c_0 x_n / a_n^\beta)$ for any fixed $\beta > 2$. The quantitative localization estimate is therefore part of the source construction.

To check that the prescribed frequency can satisfy all construction guards, consider any finite list $A\mathcal{P}_n^\nu \le k_n^\theta$ with $\theta > 0$. Choose J so that $c\nu a^2/(a-1)^3 \le \theta/2$ for all $a \ge J$ and every guard. Writing $x_n = X b_n$ with $b_n \ge J^{2n}$, the sequences a_n^2/b_n, $a_n^2 \log a_n / b_n$ and $a_n^2 \log b_n / b_n$ are bounded. Hence a single increase of X pays all the remaining logarithmic terms, uniformly in n. This proves the guards at the actual k_n. Only finitely many construction guards enter at fixed d; the fixed-m prefactors in (75)–(76) do not create new construction guards.

9.4 The smooth compact limit

Section 8 bounds the stage-n source primitives by $\mathcal{P}_n$ and verifies the prescribed localization scale for the joined construction. The preceding estimates consequently give

$$\sum_n \|\Delta u_n(0)\|_{H^m} < \infty \qquad \text{for every integer } m \geq 0.$$

For each spatial order r, take $m > r + d/2$. The legal-dimensional Sobolev embedding then gives uniform convergence of the derivative series through order r. Include the base solution and the one exceptional seed increment in $u_{\mathrm{seed},0}$. Consequently the single sum $u_0 = u_{\mathrm{seed},0} + \sum_n \Delta u_n(0)$ is smooth, has the termwise derivatives, and is divergence free. The fixed physical mean-cutoff construction puts the supports of all these genuine initial increments in one compact ball. Enlarging it to contain the base support proves $u_0 \in C^\infty_{c,\sigma}(\mathbb{R}^d)$.

All estimates concern the joined family of Section 8 at the prescribed frequencies.

10 The scale choice and the finite-stage induction

The construction uses one prescribed scale sequence. The quantitative source estimate supplies a fixed exponent D_d for the particle-map invariant and a finite list of polynomial construction guards. All choices in this section are made after those exponents have been fixed.

10.1 The base flow and the order of choices

Choose an oriented orthonormal basis (p, q, n) of E and let the full matrix L_B vanish on $E^\perp$ and satisfy

$$L_B q = p + X^{-2} n, \qquad L_B p = L_B n = 0.$$

It is trace free. Lemma 3.1, with a fixed radial cutoff, gives a compact, odd, reflection-equivariant solenoidal datum equal to $L_B x$ near zero. Its Sobolev and Gevrey bounds are uniform for $X \geq 1$. Lemma 2.1 gives its actual Euler solution u_B on a uniform short interval. Its Gevrey bounds also persist on a uniform shorter interval: with fixed base order $q_d > (d+1)/2 + 2$ and radius $r(t)$, set

$$\mathcal{X}_r = \sum_{n\geq 0} \frac{r^n}{(n!)^2} \sum_{|I|=n} \|\partial_I u_B\|_{H^{q_d}}, \qquad \mathcal{Y}_r = \sum_{n\geq 0} \frac{n r^n}{(n!)^2} \sum_{|I|=n} \|\partial_I u_B\|_{H^{q_d}}.$$

At each finite cutoff the differentiated transport cancellation and factorial product convolution give $\mathcal{X}_r' \leq C_d \mathcal{X}_r^2 + (r'/r + C_d \mathcal{X}_r/r)\mathcal{Y}_r$. Choose r' a sufficiently negative fixed constant. A short-time bootstrap bounds $\mathcal{X}_r$ and keeps $r > 0$ uniformly in the cutoff. Passing to the increasing sums proves the claim. The projected Euler equation controls the time derivative and material acceleration at a smaller radius. The integral flow equation then bounds the actual particle map and its first two time derivatives in the same manner. This proof uses the fixed legal-dimensional Sobolev embedding only. Write K_B for a uniform upper bound on its pressure Hessian.

First fix d, the packet base order and all finite inverse, coefficient, flow and source exponents, including D_d. Fix C_M, C_H large enough for the global gradient and Hessian induction below. These fix the endpoint constant and then its small threshold ζ. Choose D_0 sufficiently large for the first geometric comparison; $D_0 = 1000$ suffices for the displayed exponent 60. All seed mean-cutoff and boundary costs are fixed powers of X. Choose D_k larger than the resulting seed polynomial exponent divided by each of the finitely many frequency margins. Then choose the starting index J to pay all normal-stage exponent comparisons, and finally choose X large enough for their finite constants and total error budgets. No choice is altered during the induction.

Put $h_{J-2} = 1$, $x_{J-1} = X$, and $x_j = j^2 x_{j-1}$ for $j \geq J$. The auxiliary value h_{J-2} records the uniform scale of the base solution. Prescribe

$$h_{J-1} = X^{D_0}, \quad k_{J-1} = X^{D_k}, \quad \rho_{J-1} = X^{-D_0}, \quad \delta_{J-1} = X^{-(D_0+10)}, \tag{77}$$

$$h_j = e^{x_{j-1}/j^5}, \quad k_j = e^{x_{j-1}/j^2}, \quad \delta_j = e^{-x_{j-1}/j^3}, \quad \rho_j = e^{-x_{j-1}/j^{7/2}} \quad (j \geq J). \tag{78}$$

The seed frequency is fixed by its polynomial source bound in X; the source-envelope lemma pays the prescribed normal frequencies. Use the explicit physical mean-cutoff parameters

$$\ell_* = 1, \qquad r_* = \frac{\rho_{J-1}}{4a_d} = \frac{X^{-D_0}}{4a_d}.$$

For large X, $r_* \leq 1/4$. The splitting ball of radius $a_d r_*/\ell_*$ is exactly the seed support ball of radius $\rho_{J-1}/4$. These parameters are fixed functions of the seed value before the final choice of X; they are retained throughout the iteration and are not replaced by the shrinking ρ_j. Their reciprocals and all subsequent boundary penalties are bounded by fixed powers of X. The cost multiplier in the full mean reserve is bounded by the dimensional constant $1 + c_d 4^{-d}$, so it causes no circular change of the starting index or the source exponents.

10.2 Time and geometric comparisons

Let

$$W_j = \frac{3x_j x_{j-1}}{\sqrt{h_{j-1}}}, \qquad S_{J-1} := S_{\text{base}} = 2W_J = 6J^2 X^{2-D_0/2}, \qquad t_{J-1} = 0.$$

At stage j, the already constructed parent fixes a_j, β_j. Define

$$t_j = t_{j-1} + \frac{x_j}{\sqrt{\beta_j a_j h_{j-1}}}, \qquad S_j = t_j + 2W_{j+1}. \tag{79}$$

When $a_j, \beta_j x_{j-1}^2 \in [1/2, 2]$, $W_j/6 \leq t_j - t_{j-1} \leq 2W_j/3$. For $j > J$ the exact identities

$$\log k_{j-1} = \frac{x_{j-1}}{(j-1)^4}, \qquad \log h_{j-1} = \frac{x_{j-1}}{(j-1)^7}$$

show that W_{j+1}/W_j is arbitrarily small uniformly after J: its logarithm is

$$2\log j + 2\log(j+1) - \frac{x_{j-1}}{2j^5} + \frac{x_{j-1}}{2(j-1)^7}.$$

At $j = J$ the last term is instead $(D_0/2)\log X$. The negative term dominates in both cases. Hence $S_j \leq S_{j-1}$, and with the numerical majorant $\Theta_j = C(1 + x_j x_{j-1})$ the extra scaled interval obeys $2\sqrt{a_j h_{j-1}} W_{j+1} \leq \Theta_j^{-60}$. Also $t_J \geq S_{\text{base}}/12$, so at positive-history stages $t_{j-1}^{-1} \leq 12 S_{\text{base}}^{-1} \leq h_{j-1}$.

Use $E_* = k_{j-1}^{-1/4}$, $G_* = 1 + h_{j-2}$ and $K_h = k_{j-1}^{D_d}$. Each part of (14) satisfies its smallness guard at the same scale choice. For the spatial term the logarithm is at most

$$-\frac{x_{j-1}}{j^{7/2}} + \frac{c_d D_d x_{j-1}}{(j-1)^4} + \frac{C x_{j-1}}{(j-1)^7} + C\log x_{j-1} \leq -\frac{x_{j-1}}{2j^{7/2}}.$$

This estimate uses the inherited geometric cost, which excludes the new δ_j^{-1}; using the complete source list here would fail. The old frequency error has logarithm $-x_{j-1}/(4(j-1)^4)$. For $j \geq J+2$,

$$\log(\epsilon G_*^2) \leq -\frac{x_{j-1}}{2(j-1)^7} + \frac{2x_{j-1}}{(j-1)^2(j-2)^7} + C \leq -\frac{2x_{j-1}}{5(j-1)^7} + C.$$

For $j = J$, $\epsilon\Theta^{61} \le C(J)X^{122-D_0/2}$; for $j = J+1$ the new exponential beats the old polynomial shear. These cover the two exceptional stages. Multiplying by any of the fixed powers of x_{j-1}, Θ_j still gives the required smallness. The same comparisons give

$$h_j/h_{j-1}^2 \longrightarrow \text{arbitrarily large uniformly}, \qquad \frac{\sqrt{h_{j-1}}}{x_{j-1}x_j} \gg G_* + 1, \qquad \rho_j < r_{\text{core}}. \tag{80}$$

Here $r_{\text{core}} > 0$ is the fixed physical seed core radius. Each use of $\gg$ in this proof denotes a fixed finite inequality whose constant was chosen before J, X.

10.3 The summed budgets

The relative size estimate for the selected packet bounds its history and early-time gradient and pressure increments by

$$b_j = C_d h_j h_{j-1} K_h^{c_d} \Theta_j^{c_d} \epsilon^{-c_d} e^{-bx_{j-1}} \le e^{-bx_{j-1}/2}.$$

This follows by dividing the logarithm of the prefactor by x_{j-1}; it is bounded by $C_d/(j-1)^4 + o(1)$, with the seed predecessor treated as a polynomial in X. During the late interval the pressure sign (15), with its ambient neighborhood version, and $f'_\delta \ge -1$ give the upper-Hessian increment $C_d\delta_j h_j h_{j-1}$. Its scale satisfies

$$\delta_j h_j h_{j-1} \le e^{-x_{j-1}/(2j^3)}.$$

The exact packet remainders are bounded by $k_j^{-1/4}$. Consequently valid nonnegative increment budgets are

$$e_{p,j} = C_d(b_j + \delta_j h_j h_{j-1} + k_j^{-1/4}), \qquad e_{i,j} = C_d(b_j + k_j^{-1/4}). \tag{81}$$

The second one applies at time zero. For every fixed $A > 0$ the positive sequence x_{j-1}/j^A increases at least geometrically once J is large. Thus the sums of the displayed exponential errors, and of every fixed negative power of x_{j-1}, tend to zero with $X \to \infty$. Their finite partial sums therefore preserve both the desired upper pressure bound and the fixed-cutoff reserve of (52). This last budget includes the core term $c_d B_c r_*^d S_{\text{base}}$; it is not dropped. Initially, with the explicit frozen cutoff above, that term is

$$O_d\Big(J^2 X^{2+(1/2-d)D_0}\Big) \longrightarrow 0.$$

Indeed $B_c = O(X^{D_0})$, $r_* = O_d(X^{-D_0})$ and $S_{\text{base}} = 6J^2X^{2-D_0/2}$. This proves the assertion directly for $d \ge 3$, without an unquantified later shrink of the physical cutoff. The other reserve terms tend to zero with S_{base}. This leaves a fixed positive margin before summing the normal stages. Choosing their total $\sum_j e_{i,j} \le 1$ also gives $B_{c,j} \le B_{c,\text{seed}} + 1 = O_d(X^{D_0})$, so every later boundary penalty remains within the same polynomial seed envelope.

10.4 Seed and induction step

Set $u_{J-2} = u_B$. Apply Lemma 6.3 on $[0, S_{\text{base}}]$ with the seed scales, $m_0 = p$, $v(0) = q$, $\alpha = \delta_{J-1}h_{J-1}$ and zero mean boundary penalty. For this seed solve take $r = 0$, $B_c = 0$ and a global base lower bound B_e; the interior region is empty. Thus the required penalty is indeed $\mathcal{L} = 0$, and the seed inverse is coercive on the short interval without a special core term. At time zero $m \cdot M_B v = 1$ throughout its compact packet support. The uniformly bounded base equations preserve $m \cdot M_B v \ge 1/2$ on the chosen short interval. Thus the seed upper-pressure cost is at most $C_d X^{-10} + k_{J-1}^{-1/4}$. At $t_0 = 0$ every forced high and mean grade has zero initial value; only the primary and its actual tensor corrector remain. They are supported in the fixed seed core. The interior gradient bound is $B_c = C_M h_{J-1} + 2$; the exterior lower bound remains that of the base. Equip this actual seed output with the already prescribed fixed cutoff parameters and penalty $\mathcal{L} = C_{1,d} B_c$ for subsequent stages. Their full reserve is strictly small by the explicit estimate above.

The seed expansion is, at the fixed origin,

$$M_{J-1} = M_B + h(t)q(t)p(t)^T + \mathcal{E}(t), \qquad |\mathcal{E}| \le k_{J-1}^{-1/4}, \qquad h(0) = h_{J-1}.$$

The normalized frame is transported by the actual base gradient. Its activation parameters are exactly $a_J = 1$ and $\beta_J = X^{-2}$. The actual lifted flow estimate gives its polynomial particle-map bound at k_{J-1}. Thus the complete input state for stage J consists of two exact solutions, the actual frame and shear decomposition, the global low bounds, the strict fixed-cutoff reserve and the polynomial label bounds. The first activation time is zero, so this stage uses the forward primary without a history interval.

Assume that state at stage j. Keep the actual parent on its available long interval $[0, S_{j-1}]$ while forming the endpoint and geometric source. Its actual scaled horizon is $a_j(S_{j-1} - t_{j-1})/\epsilon_j$, bounded by the numerical majorant Θ_j. By $S_{j-1} - t_{j-1} = 2W_j$, it contains the target and the short retained extension, and the scalar comparison interval lies within these bounds. The numerical majorant is not an assertion that coefficient paths have been defined beyond their actual time interval. Only after these source estimates and the target normalization are obtained do we restrict the parent and coefficient fields to $[0, S_j]$ for the packet and correction. Their coefficient constants remain those of the long parent. All Duhamel and energy estimates may be integrated on the actual scaled interval of length at most Θ_j; no extension to the numerical majorant is needed. The relative comparison with the target size is used only through S_j, whose extension past the target is small, and is not asserted on the whole longer parent interval. Along the older central solution, $\dot{B} = -B^2 - H_{j-2}$, which gives the bounds used above on $B, \dot{B}$. Lemma 5.1 supplies the actual stationary history for $j > J$, and the zero-time prescription supplies it for $j = J$. The ambient comparison and source-envelope argument verify the quantitative neighborhood, full transverse propagator and every coefficient cost at the already prescribed ρ_j, k_j. Take

$$\alpha_j = \frac{\delta_j h_j}{|m(t_j, 0)||v(t_j, 0)|}.$$

The joined construction in Lemma 6.3 then gives an exact smooth odd reflection-equivariant solution u_j on $[0, S_j]$. Its full ambient tensor correctors, all means and the residual correction are part of this same solution. The graph-flow identities give the next actual particle-map invariant $K_{h,\text{next}} \le k_j^{D_d}$.

The global expansion gives

$$\|M_j - M_{j-1}\|_\infty \le C_d h_j + k_j^{-1/4}, \qquad \|H_j - H_{j-1}\|_\infty \le C_d h_j h_{j-1} + k_j^{-1/4}.$$

With (80) these preserve $\|M_j\|_\infty \le C_M h_j$ and $\|H_j\|_\infty \le C_H h_j h_{j-1}$. The one-sided budgets (81) preserve the pressure upper bound, the initial interior and exterior quadratic-form bounds, and the full fixed-cutoff reserve. No locality of the pressure is being used: its signed expansion and its error bound hold globally.

For $t \in [t_j, S_j]$, the new normalized pair p_2, q_2 satisfies

$$M_j = M_{j-1} + h(t) q_2 p_2^T + \mathcal{E}_j, \qquad |\mathcal{E}_j| \le k_j^{-1/4}, \qquad h(t_j) = h_j.$$

Its frame equations have older field M_{j-1}, by the exact ray and velocity equations, and $h'/h = -(M_{j-1})_{p_2 p_2} - (M_{j-1})_{q_2 q_2}$. Lemma 5.3 bounds each relative change in a_j by $C_d x_{j-1}^{-4}$, whose sum is arbitrarily small. Starting at one, the product stays in $[1/2, 2]$. The same lemma gives directly $\beta_{j+1}x_j^2 \in [1/2, 2]$; that estimate is not multiplied across stages. Its compression bound and (80) give $p_2 \cdot M_{j-1}p_2 < -1$, while $|\mathcal{E}_j| < 1$. Also $(C_M h_{j-1} + 1)/h_j < \zeta$. These prove the two actual activation conditions for the next endpoint problem. The low bounds on the entire history and $t_j^{-1} \le h_j$ prove its history hypotheses. These estimates establish every component of the induction state for the constructed child.

11 The limiting datum and endpoint

The initial estimates of Section 9, applied to the same joined finite families and the actual source envelope, give

$$\sum_{j \ge J} \|u_j(0) - u_{j-1}(0)\|_{H^m} < \infty \qquad \text{for every fixed } m.$$

The seed is one fixed smooth increment. Lemma 7.2 places all these initial velocities in one compact set. Sobolev embedding at arbitrarily high fixed orders therefore proves that their single limit u_0 is smooth, compactly supported and divergence free.

The target times increase to a finite positive $T_\infty \le S_{\mathrm{base}}$. At the central target the exact expansion gives

$$|\nabla u_j(t_j, 0)|_{\mathrm{F}} \ge h_j - C_M h_{j-1} - k_j^{-1/4} \longrightarrow \infty.$$

Let u be the unique maximal smooth Sobolev solution from u_0. If its lifetime T^* exceeded T_∞, it would be smooth on a closed interval containing all t_j, with bounded H^{s+1} norm. Apply Lemma 2.3 on the actual varying intervals $[0, t_j]$ to u_j and u. Their initial H^s difference tends to zero, so their gradients converge uniformly on those intervals. The displayed divergence of the central gradients would contradict boundedness of ∇u there. Thus $0 < T^* \le T_\infty < \infty$. This argument does not assert equality of T^* and T_∞. Lemma 2.4 now supplies the exact Frobenius gradient limsup and the once-counted antisymmetric-vorticity integral in H_d. Lemma 2.2 fixes the literal pressure normalization, and Lemma 2.1 supplies every required Sobolev order on each shorter closed time interval.

The quantitative joined-source estimate and the prescribed scale sequence verify the hypotheses at every finite stage. The initial limit, finite lifetime and continuation conclusions therefore prove Theorem 1.1, with no additional assumption on the dimension or on a prospective solution.

Disclosure

All derivations in this article were completed by artificial intelligence.

References

[1] OpenAI, *Finite time blowup for the Euler equation*, manuscript, `https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf`.